Resistive switching and memory computing

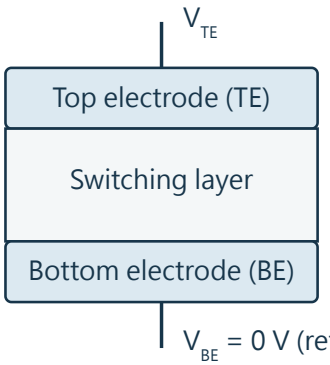
A resistive memory cell typically has a top electrode, switching layer and bottom electrode. A programming pulse changes its stored state; a specified read bias probes its current.

LRS and HRS mean low and high resistance. Assigning either state to a binary value is a system convention, not a material law. We connect measurements to physical models, arrays and reliability without treating one observation as proof of a unique mechanism.

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Cell structure and voltage reference

Two-terminal cell: define voltage first



Define $V = V_{TE} - V_{BE}$

Operations and readout

SET: higher conductance

RESET: lower conductance

Read: state voltage and duration

$G_{read} = I_{read} / V_{read}$

Polarity and current limit are device-specific; no particular mechanism is assumed.

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Read conductance has measurement conditions

At a specified voltage and time, define $G_{read} = I_{read}/V_{read}$. This ratio is an apparent conductance at that operating point.

Nonlinear response, polarity asymmetry or drift can make it different at another bias or time. A read can also disturb the stored state.

A voltage sweep traces a response under a specified sweep rate and current limit. Pulse experiments specify amplitude, duration and sequence and can probe switching and read disturbance.

Report temperature, voltage polarity, compliance, pulse timing and read protocol. A single curve does not establish device-to-device or cycle-to-cycle distributions.

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Forming, SET, RESET and read

- Forming is an initial conditioning step required by some devices, not all.
- SET generally moves to a more conductive state; RESET moves to a less conductive state.
- Read probes the state. Current compliance constrains programming current but does not guarantee freedom from damage.

For bipolar switching, SET and RESET generally use opposite bias polarities. Unipolar switching can use the same polarity with different pulse or current-limit conditions.

Define voltage explicitly, for example $V = V_{top} - V_{bottom}$, before naming positive or negative. A polarity observed in one stack is not universal.

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Trace a pulse protocol

This synthetic example assumes a particular bipolar stack and $V = V_{top} - V_{bottom}$; the numbers are illustrative, not specifications. Record the read immediately after each pulse and again later to detect drift.

Step	Applied pulse	Current limit	Read at 0.1 V	Inference
Initial read	+0.1 V, 10 µs	read circuit	1 µA	$G_{read} = 10$ µS
SET	+1.0 V, 1 µs	100 µA	20 µA	$G_{read} = 200$ µS
RESET	−1.0 V, 1 µs	not specified	1 µA	$G_{read} = 10$ µS

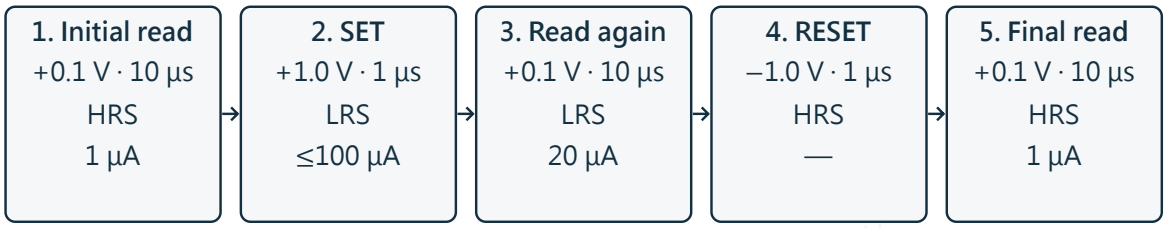
Compliance constrains programming current; a low read pulse is not automatically nondestructive. Verify repeated reads and record drift instead of inferring it from one measurement.

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Pulse operations and state sequence

Pulse and read sequence: illustrative example

Example only; V = Vtop − Vbottom.



Read currents shown; SET limited; RESET current unspecified.
Not to scale; reads assumed nondisturbing.

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ECM: a possible redox pathway

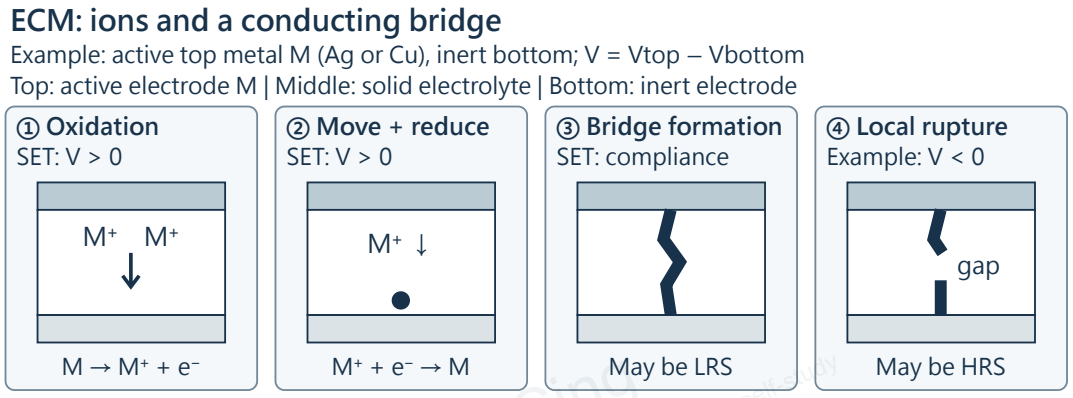
In electrochemical metallization (ECM), an active electrode can oxidize; metal cations migrate through the switching layer and reduction can form a conductive bridge. Subsequent operation can narrow or rupture it.

For a hypothetical Ag / oxide / inert-electrode stack with Ag positive during SET, oxidation $\mathrm{Ag} \rightarrow \mathrm{Ag}^{+} + e^{-}$ supplies cations while electrons enter the external circuit. Reduction $\mathrm{Ag}^{+} + e^{-} \rightarrow \mathrm{Ag}$ can deposit metal where electrons are available.

Nucleation, growth direction and RESET depend on electron supply, transport and reaction kinetics. This polarity and schematic are examples, not universal rules.

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ECM stages for one electrode configuration



Schematic, not to scale; growth direction depends on materials and reaction kinetics.
RESET polarity, rupture location and pulse depend on device; specify read conditions.

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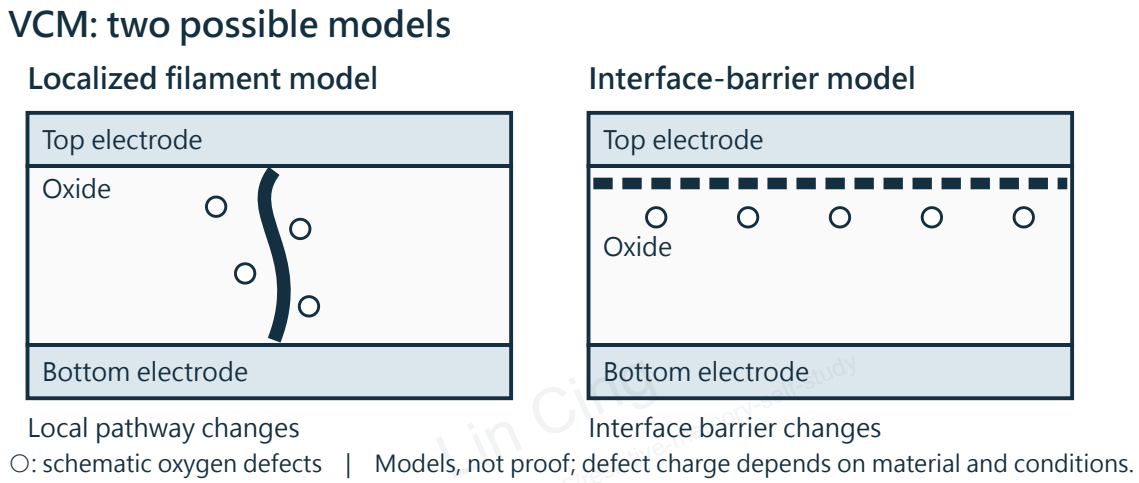
VCM and competing explanations

Valence-change mechanisms (VCM) involve oxygen-ion or oxygen-vacancy transport and local redox changes. Redistribution changes local stoichiometry and electronic transport. One model emphasizes a filament; another emphasizes an interface barrier. Each requires specific electrode, field and redox assumptions. Thermochemical contributions and unipolar behavior are not universally equivalent.

A hysteretic I–V curve is not proof of a unique mechanism. Local chemical maps, conductive microscopy, in-situ imaging and time-resolved measurements can test models, subject to resolution and possible measurement perturbations; correlation alone does not establish causality.

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Two candidate VCM models



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Model a state without assuming a mechanism

One framework is $i(t) = G(x(t), v(t), t)v(t)$ and $\dot{x} = f(x, v, t)$. The chosen state x may be multidimensional, and G need not be independent of voltage or time.

An ideal memristor is not synonymous with every two-terminal device displaying hysteresis. A model must state its domain and be tested against measurements.

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An ideal crossbar computes column currents

Let row i be driven at V_i and the conductance at row i, column j be G_{ij}. Assume linear read response, negligible wire resistance, and columns held at virtual ground.

Each branch carries $I_{ij} = G_{ij}V_i$. Kirchhoff's current law then gives $I_j = \sum_i G_{ij}V_i$. These assumptions are part of the result.

For $G = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}\ \mu\mathrm{S}$ and $V = (0.2, 0.1)\ \mathrm{V}$, the two column currents are $I_1 = 2(0.2) + 1(0.1) = 0.5\,\mu\mathrm{A}$ and $I_2 = 1(0.2) + 3(0.1) = 0.5\,\mu\mathrm{A}$. Here $\mu\mathrm{S} \cdot \mathrm{V} = \mu\mathrm{A}$; the numbers illustrate the model.

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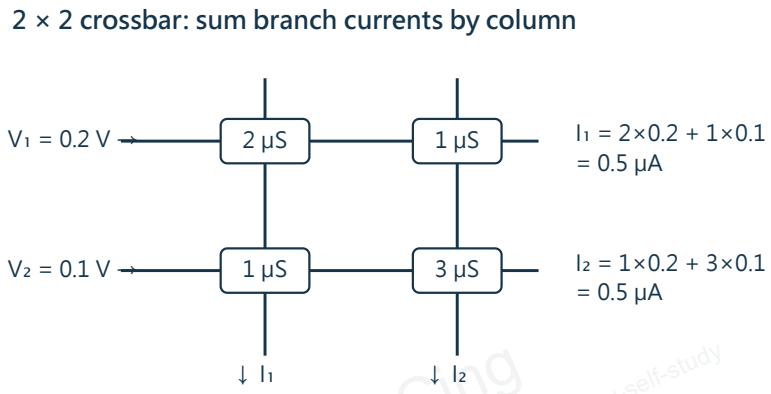
Derive the array sum at the cell terminals

Let the local row and column potentials at cell (i, j) be v_{ij} and u_{ij}. For a linear read conductance, the branch current into the column is $I_{ij} = G_{ij}(v_{ij} - u_{ij})$. Kirchhoff's current law sums these branch currents at the column sensing node. Only when row wires are ideal and the column is held at virtual ground do $v_{ij} = V_i$ and $u_{ij} = 0$ yield $I_j = \sum_i G_{ij} V_i$.

With the earlier 2×2 example, suppose both local row-2 potentials are 0.08 V, row 1 remains 0.2 V, and column potentials remain zero. Then $I_1 = 2(0.2) + 1(0.08) = 0.48$ µA and $I_2 = 1(0.2) + 3(0.08) = 0.44$ µA. This assumes the local voltages are known; solving a real resistive network determines them rather than prescribing them.

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Ideal crossbar: two input rows



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When the array differs from the model

Wire resistance changes local voltages; finite sensing impedance changes column potential. Leakage, unintended paths, nonlinear response, conductance drift and programming error can change the measured sum.

A negative weight needs an explicit representation, such as a difference of conductances, and corresponding sensing circuitry. A single passive conductance is not a negative weight.

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Reliability is a measurement, not a slogan

Retention describes distinguishability after a stated time and temperature, with a defined read and failure criterion. Endurance is a cycle count under stated waveforms and failure criterion.

Separate cycle-to-cycle from device-to-device variation. Read margin depends on the sensing circuit and distributions, not merely two nominal resistance values.

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Compare reliability only under matching protocols

Consider two hypothetical measurements, each on 100 cells. Test A reports 95 cells retaining distinguishable read currents after 1 hour at 85 °C; test B reports 98 cells after 1 hour at 25 °C. Both must specify read voltage and time, current thresholds, prior cycling and whether failed cells were excluded. Even with those records, different temperatures prevent treating the observed percentages as a controlled comparison of two materials.

For endurance, record the SET/RESET amplitudes, pulse widths, verify procedure and failure criterion. A short high-temperature retention measurement alone cannot establish 10-year retention at room temperature; an extrapolation requires a supported model, uncertainty and appropriate validation.

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Applications and device requirements

Four uses: operations and device demands

Storage	Write state → read and decide Focus: read margin, retention, write endurance
Stateful logic	Read state → update state Focus: operations may consume writes
Offline inference	Program weights → repeated reads Focus: read accuracy, drift and converters
Online training	Read and compute → update weights Focus: update error, asymmetry and endurance

Conceptual flows; operations and peripheral costs depend on architecture.

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Storage and selectors: the array sets the constraints

For digital storage, a read circuit must distinguish the programmed states at specified voltages and times. A large single-cell resistance ratio does not by itself guarantee a usable array read margin: wire resistance, sensing noise and currents through unselected cells also matter.

An access device can suppress unselected-path currents using gate control, as in 1T1R, or using a nonlinear two-terminal selector. Its leakage, drive current, switching speed and integration cost must be assessed together with the memory element; a selector is not automatically an ideal switch.

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Stateful logic can require repeated writes

In stateful logic, a programmed device state participates in a logic operation and the result is stored by a subsequent state change. The precise voltage sequence and number of writes depend on the circuit and its logic convention. A reconfigurable circuit instead may write a configuration once and use it for many operations; do not assume each operation rewrites its configuration.

Suppose the **same cell** receives exactly k writes per operation, operations occur at N per second, and this rate is constant for T seconds. The cell accumulates NkT writes. For $k = 2$, $N = 10^3\,\mathrm{s}^{-1}$ and $T = 60\,\mathrm{s}$, this is 1.2×10^5 writes. If writes are spread across cells, sum each cell's actual writes instead. A write is not necessarily a complete SET/RESET cycle; endurance depends on the specified pulses and failure criterion.

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Inference and training stress different properties

An analog crossbar can use programmed conductances to perform an approximate weighted sum. Offline inference programs weights before use: read noise, conductance drift, line resistance and ADC/DAC precision affect the result. Weight signs need an explicit representation, such as a differential pair.

Online training also updates conductances during learning. Update nonlinearity, asymmetry, quantization, stochasticity and write endurance can affect the training trajectory. Impact depends on the model, workload, mapping and compensation; neither inference nor training is automatically tolerant of every device imperfection.

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Temporal randomness differs from a device fingerprint

A stochastic switching device may produce different outcomes across repeated identical pulses. A random-number generator uses temporal variation; bias, serial correlation, readout and any post-processing must be characterized. Repeated switching also consumes endurance.

A physical unclonable function (PUF) instead aims to extract a reproducible device-specific response from spatial differences among devices. Its usefulness depends on stability under environmental changes, uniqueness across devices and measurement noise. Temporal randomness and spatial variation are different properties, even if a single device technology exhibits both.

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Phase-change memory: the thermal pathway

In a typical phase-change memory (PCM), a heater raises a chalcogenide region's temperature. A sufficiently long heating interval can allow crystal nucleation and growth, producing a more conductive SET state. A RESET pulse heats an active region above its melting range; rapid cooling can freeze a more resistive amorphous region.

These are illustrative thermal routes, not universal pulse recipes: nucleation and growth rates, actual temperature distribution, quench rate and phase history matter. Crossing a nominal temperature alone is not a guarantee of phase change. ECM and VCM can involve redox and ion transport as well as heating; a similar electrical hysteresis does not make their microscopic processes identical to PCM.

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A lumped thermal model

Let T_0 be ambient temperature, C_{th} thermal capacitance, R_{th} thermal resistance and P constant power during a pulse. The energy balance is $C_{th}\,dT/dt = P - (T - T_0)/R_{th}$, with $T(0) = T_0$. Its solution is $T(t) = T_0 + PR_{th}(1 - e^{-t/\tau})$, where $\tau = R_{th}C_{th}$.

This model assumes one uniform temperature, constant parameters and a known power waveform. A real pulse need not deliver constant power: resistance changes during switching and spatial hot spots can invalidate the single-node approximation.

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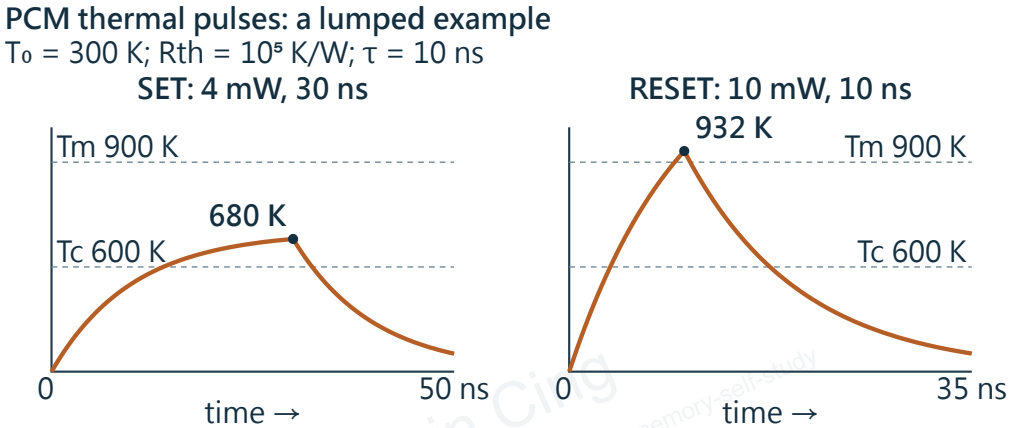
PCM pulse example: calculate before naming a state

Assume $T_0 = 300$ K, $R_{th} = 10^5$ K/W and $\tau = 10$ ns. A 4 mW pulse for 30 ns gives $T = 300 + 400(1 - e^{-3}) = 680.1$ K. In an independent trial starting at 300 K, a 10 mW pulse for 10 ns gives $T = 300 + 1000(1 - e^{-1}) = 932.1$ K.

If illustrative crystallization and melting thresholds are 600 K and 900 K, respectively, the first peak lies between them and the second exceeds the melting threshold. These peaks alone do **not** prove SET or RESET: kinetics and cooling remain necessary to assess.

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Heating and quenching are separate stages



Assumed thresholds; temperature alone does not prove a phase change.

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Cooling after a RESET pulse

For the 932.1 K endpoint, set $P = 0$ after the pulse and keep T_0 and τ constant. Cooling follows $T(t) = 300 + (932.1 - 300)e^{-t/(10\,\mathrm{ns})}$; reaching 600 K takes $10\ln(632.1/300) = 7.45$ ns. This is a modelled cooling time, not proof that a particular quench rate yields a stable amorphous state.

The initial cooling rate is $dT/dt = -(932.1 - 300)/(10\,\mathrm{ns}) = -63.21$ K/ns. It slows as temperature approaches the ambient value. A spatial temperature profile or a change in thermal parameters can alter both the cooling time and the locally realized phase.

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A read-and-write OR: make the circuit boundary explicit

Consider two memory cells connected in parallel to a 0.1 V read supply and a virtual-ground current sensor. A stored 0 is $1\,\mu\mathrm{S}$, and a stored 1 is $10\,\mu\mathrm{S}$. Assume linear, stable, non-disturbing reads, ideal wiring and negligible sensor error. Then $I = 0.1(G_A + G_B)$ in μA.

A	B	Sum current (µA)	$I > 0.6\,\mu$A?
0	0	0.2	0
0	1	1.1	1
1	0	1.1	1
1	1	2.0	1

The comparator computes OR under the stated convention; it does not itself write a memory cell.

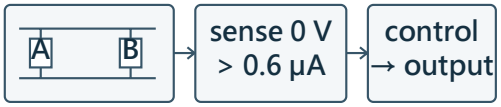
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Read-and-write OR and spike timing

Parallel OR read and synaptic pulses

1. Read, compare, then write

V = 0.1 V; 0/1 = 1/10 µS

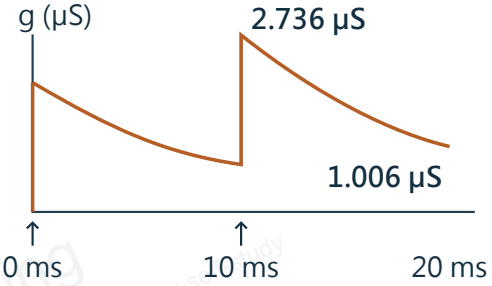


A	B	I (µA)	OR
0	0	0.2	0
0	1	1.1	1
1	0	1.1	1
1	1	2.0	1

Hypothetical model; control writes a separate preset output.

2. Two synaptic pulses

Δg = 2 µS; τ = 10 ms



Ideal pulses; exponential decay.

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From the comparator to a stored result

To store the OR result, first initialize a separate output cell to HRS. The digital controller applies the appropriate SET pulse only for comparator output 1 and verifies the result. This example uses sensing and digital control; it is not evidence of an autonomous native stateful gate. Cell variation, sensing margin, leakage, read disturbance and write failures can change the outcome.

For the assumed currents, the low-state current is 0.2 µA and the smallest high-state current is 1.1 µA. With an ideal threshold of 0.6 µA, their distances from the threshold are 0.4 and 0.5 µA. These are nominal margins, not an error probability; noise and distributions must be measured.

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A dynamic synapse: events and decay

As an illustrative short-term model, between spikes $dg/dt = -g/\tau$ and each instantaneous spike adds Δg. This is a chosen dynamical model, not a claim that every resistive device follows exponential decay. Let $g(0^-) = 0$, $\tau = 10$ ms and $\Delta g = 2\,\mu$S, with spikes at 0 and 10 ms.

After the first spike $g(0^+) = 2\,\mu$S. Just before the second, $g(10^-) = 2e^{-1} = 0.7358\,\mu$S; just after it, $g(10^+) = 2(1 + e^{-1}) = 2.7358\,\mu$S. At 20 ms without another spike, $g(20) = 2.7358e^{-1} = 1.0064\,\mu$S. At a non-disturbing 0.1 V read, $I(10^+) = 0.27358\,\mu$A.

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Routing spikes is part of the computation

An event can carry a source address and timestamp through local and inter-core routers; a routing table maps the event to destination synapses. Multicast sends one event to several destinations, while programmed connectivity decides which synapses receive it. The dynamic update above must be evaluated at the actual arrival time.

For example, an event generated at 0 ms and delayed by 2 ms reaches a synapse at 2 ms; the next arrives at 12 ms if it has the same delay. Their separation remains 10 ms. Unequal delays, packet loss and congestion can change the effective spacing, so circuit behavior cannot be inferred from the source timestamps alone.

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Random switching: a probability model and its limits

If a specified experiment is modelled by a constant switching hazard λ, the waiting time is exponential: $P(T \le t) = 1 - e^{-\lambda t}$. Thus $\lambda = 10^6\,\mathrm{s}^{-1}$ and $t = 1\,\mu$s give $P(T \le t) = 1 - e^{-1} \approx 0.632$. A constant hazard, reset protocol, voltage and temperature are assumptions to test, not universal device laws.

For raw independent, identically distributed bits with $P(1) = 0.6$, min-entropy is $H_{min} = -\log_2(0.6) = 0.737$ bit per raw bit. This one-bit statistic alone does not measure entropy rate if bits are correlated or the distribution changes over time.

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Debiasing is conditional on the source

Under the independent, identically distributed model, nonoverlapping pairs 01 and 10 each have probability $0.4 \times 0.6 = 0.24$. Map 01 to 0 and 10 to 1; discard 00 and 11. The acceptance probability is 0.48 per pair, giving 0.24 output bits per raw bit on average, with equal probabilities for accepted 0 and 1.

Correlation, drift or selection bias can break that guarantee. Characterize the source, dependence, operating conditions and post-processing; passing a finite collection of statistical tests does not by itself prove cryptographic unpredictability.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Arrays, selectors and three-dimensional integration

A single RRAM cell must become part of an addressable network. This chapter follows the current through that network, then examines access devices, sensing and the physical layout.

All numerical examples are deliberately simplified circuit models, not measured device specifications.

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A crosspoint is a cell, not an isolation switch

At the intersection of row R_i and column C_j, a memory element connects the two lines. Applying a voltage to a selected row and sensing a column also establishes voltages across other intersections.

Those other cells can carry currents into the sense circuit, or form a path through floating lines. Which currents appear depends on the line-bias and sensing circuit; a cell's HRS/LRS ratio alone does not determine array performance.

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Fixed $V/2$ bias: define every line

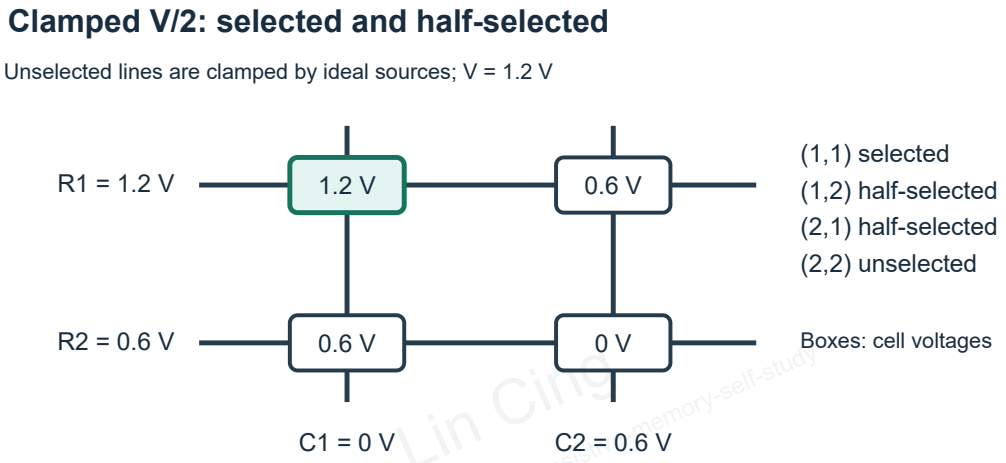
Drive the selected row to V, the selected column to 0 and every unselected row and column to $V/2$. Assume ideal voltage sources and wires.

Cell position	Row potential	Column potential	Row − column
Selected intersection	V	0	V
Selected row only	V	$V/2$	$V/2$
Selected column only	$V/2$	0	$V/2$
Neither selected	$V/2$	$V/2$	0

For $V = 1.2$ V the half-selected cells see 0.6 V. This is one bias scheme; finite line resistance or voltage division within a 1S1R cell changes the device-level voltages.

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Fixed-bias example: follow the sources



The unselected lines are **clamped**, not floating: their sources can supply or absorb current. The selected-column current can contain both the selected-cell current and current from a half-selected cell.

Half-select is a stress condition

An unselected cell experiences a fraction of the applied pulse. Whether it remains unchanged depends on its initial state, device voltage, polarity, pulse width and number of repetitions—not on voltage alone.

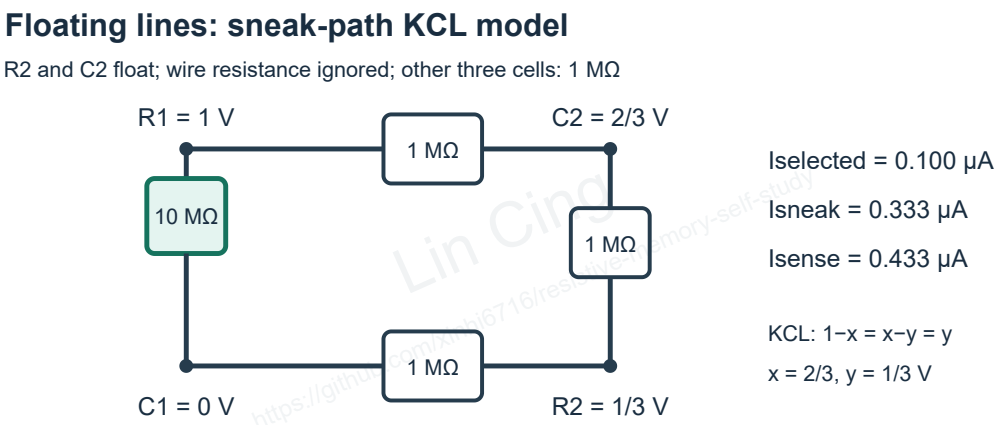
For an array, count the pulses imposed on each half-selected cell by its neighbors' writes. Compare its resulting distribution against the read criterion, including variability and cumulative disturb.

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Floating lines create a different network

Take a 2×2 passive array: row $R_1 = 1$ V and column $C_1 = 0$ V; R_2 and C_2 float. The selected cell R_1–C_1 has conductance $0.1\,\mu$S; each other cell has $1\,\mu$S.

Assume linear, fixed-state conductances, ideal wires and no connection from a floating line to an external load. A path $R_1 \to C_2 \to R_2 \to C_1$ now runs in parallel with the selected cell.



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Solve the floating network by KCL

Let the unknown potentials of C_2 and R_2 be x and y volts. With all three unselected conductances equal to $1\,\mu$S, Kirchhoff's current law gives

$$(x-1)+(x-y)=0, \qquad (y-x)+y=0.$$

Thus $2x-y=1$ and $2y=x$, so $x=2/3$ V and $y=1/3$ V. Each of the three series branches drops $1/3$ V.

Selected-cell current is $0.1\,\mu$A; the alternative path carries $1/3\,\mu$A. The sense node at C_1 receives $0.1+1/3=0.433\,\mu$A, approximately.

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Compare the boundary conditions

In fixed $V/2$ bias the other lines are held by sources; in the floating example their voltages are **unknowns determined by KCL**. The three-resistor path calculation applies to that floating 2×2 topology, not automatically to a clamped array.

An HRS selected cell surrounded by LRS cells is an illustrative adverse pattern: the unwanted current can exceed the selected current. The actual worst pattern depends on the complete bias, device and sensing model.

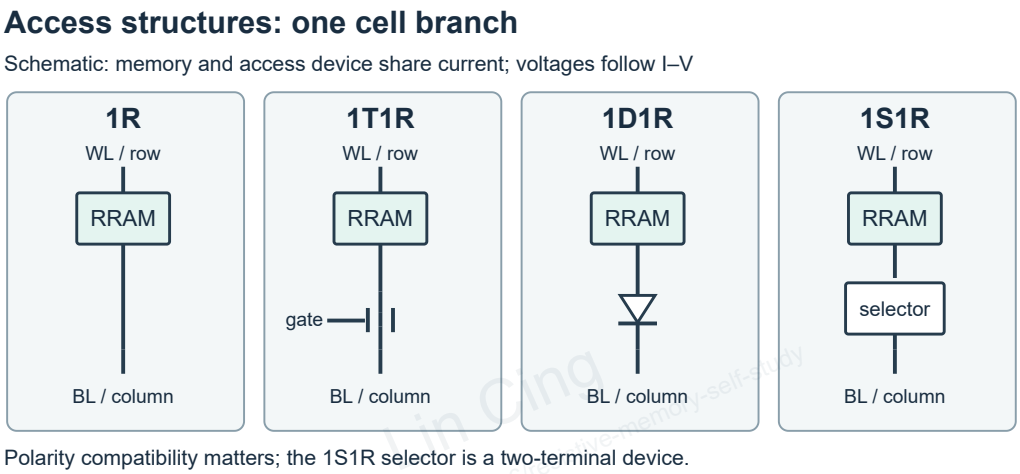
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Access structures change the available controls

Structure	Access mechanism	Design consequence
1R	Memory element alone	Isolation relies on its own I–V and bias scheme.
1T1R	A gated transistor in series	Gate control can suppress leakage and limit current, at a footprint and voltage cost.
1S1R	Two-terminal selector in series	Selector nonlinearity suppresses some low-bias currents without a gate.
1D1R	Diode in series	Rectification favors one direction; compatibility with both programming polarities must be checked.

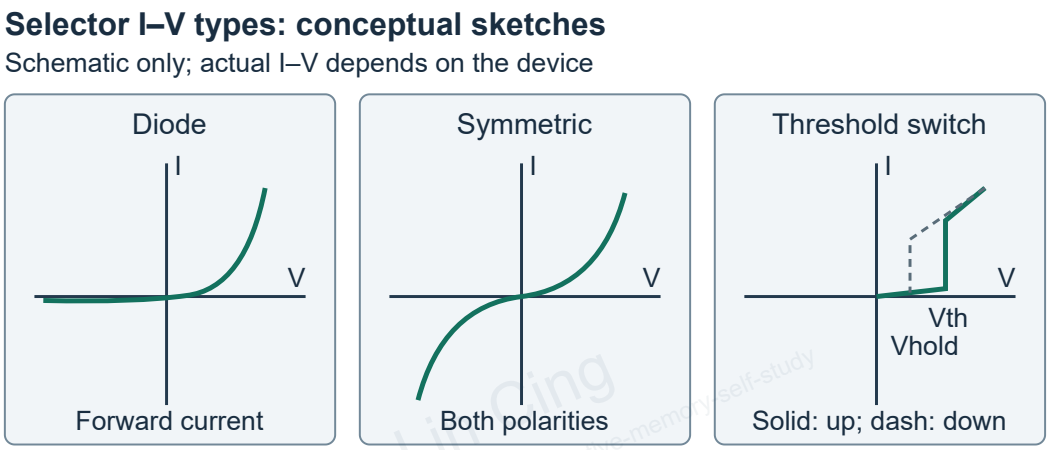
A self-selective element incorporates access nonlinearity into its stack; it does not provide the separate gate of 1T1R.

Access structures: follow each control path



The transistor has a separate gate terminal. A two-terminal selector or diode is driven through the series path; the memory and its access element share the applied voltage.

Three selector response families



A **rectifier** conducts asymmetrically in the two polarities. A **varistor** gives a strongly nonlinear response; it may be engineered for conduction in both directions, but symmetry is not automatic. A **threshold switch** can enter a conducting state after a threshold and return when its holding condition is lost.

Each label describes a response family, not a guarantee of leakage, switching speed, retention or suitability for a given memory.

Threshold and holding conditions

For a hypothetical volatile selector, an increasing voltage reaches a turn-on threshold V_{th}; after turn-on, the current and device voltage follow the selector's conducting branch. The selector returns to its off branch when current falls below a holding current I_h in the stated model.

The conducting branch can have hysteresis, and real thresholds depend on polarity, pulse duration and history. A threshold selector is not an ideal permanent latch and must not be confused with the nonvolatile state stored in RRAM.

The source, wire and series memory constrain the trajectory: crossing an isolated-device threshold alone does not prove a stable operating point.

A series selector shares voltage with memory

The same current flows through both devices and their voltages satisfy $V_{app} = V_S + V_M$. For nonlinear devices, solve $I_S(V_S) = I_M(V_M)$ together with that sum, using the relevant state, polarity and branch.

For a **fixed-state linear illustration**, let $R_S = 2\ \text{k}\Omega$, $R_M = 4\ \text{k}\Omega$ and $V_{app} = 1.2\ \text{V}$. Then $I = 1.2/6\,\text{k}\Omega = 0.2\ \text{mA}$; $V_S = 0.4\ \text{V}$ and $V_M = 0.8\ \text{V}$. Neither element receives the full applied voltage.

For a threshold selector use its actual I–V branches and circuit load line; fixed-resistance division cannot predict the switching transient or the holding condition.

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A nonlinear operating point: a piecewise model

Assume a positive-bias, non-hysteretic selector with $I = V_S/(100\,\text{k}\Omega)$ for $V_S < 0.8\ \text{V}$, and $I = (V_S - 0.8\ \text{V})/(1\,\text{k}\Omega) + 8\ \mu\text{A}$ for $V_S \geq 0.8\ \text{V}$. A fixed $4\,\text{k}\Omega$ memory is in series; $V_{app} = 1.2\ \text{V}$.

Try the low-voltage branch: $I = 1.2\ \text{V}/(104\,\text{k}\Omega) = 11.54\ \mu\text{A}$ and $V_S = 1.154\ \text{V}$. That violates $V_S < 0.8\ \text{V}$, so reject it.

On the high branch, $V_S = 0.8\ \text{V} + (1\,\text{k}\Omega)(I - 8\ \mu\text{A})$ and $1.2\ \text{V} = V_S + (4\,\text{k}\Omega)I$. The intersection is $I = 81.6\ \mu\text{A}$, $V_S = 0.8736\ \text{V}$ and $V_M = 0.3264\ \text{V}$, consistent with the branch. This is a teaching model, not a measured selector.

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Check the read and write windows separately

At the intended read voltage, distinguish HRS and LRS while limiting leakage and avoiding read disturb. At both SET and RESET polarities, the combined stack must deliver the memory's required voltage, current and pulse duration.

Also check selector on-current, half-select leakage, endurance, temperature, speed and variability. A selector that blocks reverse current may be suitable for a particular unipolar scheme but conflict with bipolar operation; topology alone cannot establish compatibility.

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What a nonlinearity ratio tells us

For a specified device state, polarity and measurement timing, define

$$NL = \frac{|I(V_{read})|}{|I(V_{read}/2)|}.$$

The denominator must be nonzero, and numerator and denominator must use the same current sign or magnitudes consistently. For $V_{read} = 1.2\,\mathrm{V}$, $NL = 10^3$ and $|I(1.2\,\mathrm{V})| = 10\,\mu\mathrm{A}$, the model gives $|I(0.6\,\mathrm{V})| = 10$ nA.

This synthetic example compares only two points. It does not specify other voltages, switching dynamics, noise or complete-array read margin.

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Read margin needs a sensing definition

Let I_H and I_L denote the sensed currents for a selected HRS and LRS, including all paths admitted by the chosen bias circuit. In a simple current-threshold reader, choose T and call the state LRS if $I_{sense} > T$.

If the possible HRS currents lie in $[H_{min}, H_{max}]$ and LRS in $[L_{min}, L_{max}]$, a deterministic threshold separating both ranges exists only if $H_{max} < L_{min}$. The separation $L_{min} - H_{max}$ is a current-domain margin under this definition.

Changing the sense circuit or using a voltage-domain margin changes the calculation; noise and distributions must be included before claiming an error rate.

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A small current-margin calculation

Use the previous floating 2×2 network, assume the three unselected cells remain linear at $1\,\mu$S, and switch only the selected conductance between 0.1 and $1\,\mu$S. Both states see 1 V across the selected cell and have the same $1/3\,\mu$A alternative-path current.

Hence $I_H = 0.433\,\mu$A and $I_L = 1.333\,\mu$A. With threshold $T = 0.8\,\mu$A, the HRS and LRS separations from threshold are 0.367 and $0.533\,\mu$A respectively. These nominal values assume an ideal current sense at 0 V, no noise and no state variation.

If unselected states or node voltages vary between reads, recalculate both ranges. The example does not establish how margin scales for a larger array.

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Threshold window with an assumed background

For a separate example, hold the sense column at 0 V and assume the sneak contribution is fixed at $0.3\,\mu$A regardless of the selected state. Let the selected HRS and LRS currents be 0.1 and $1.0\,\mu$A, respectively. The sensed currents are then $I_H = 0.4\,\mu$A and $I_L = 1.3\,\mu$A.

With the rule "LRS if current is strictly greater than T," both nominal states are classified correctly when $0.4 \le T < 1.3\,\mu$A. This definition gives a current window $I_L - I_H = 0.9\,\mu$A; other definitions may normalize the margin differently.

The $0.3\,\mu$A is an explicit simplifying assumption, not the $1/3\,\mu$A computed for the earlier floating network. In general solve the entire network separately for HRS and LRS.

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Add variability and bounded sensing error

In a particular operating condition, suppose measured HRS currents lie in $[0.35, 0.55]\,\mu$A and LRS currents in $[1.10, 1.40]\,\mu$A. Let the sensing error have absolute value at most $0.05\,\mu$A. The resulting observed ranges are $[0.30, 0.60]$ and $[1.05, 1.45]\,\mu$A.

Using the same strict-greater-than threshold rule, every value in these bounds is classified correctly for $0.60 \le T < 1.05\,\mu$A. If the ranges overlap, no single threshold guarantees correct classification for all bounded cases.

This is a bounded-error example. Actual distributions, sample sizes, correlations and failure rates need separate evaluation.

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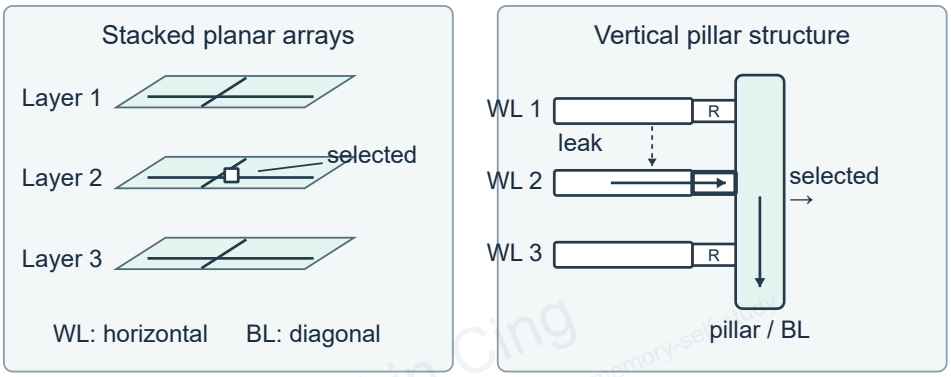
Array size: where leakage accumulates

Larger arrays generally introduce more possible paths and longer wires, but the actual sensed leakage depends on topology, biases and nonlinear I–V characteristics. A sum of identical half-select currents is valid only if the assumed circuit really presents those same device voltages and currents.

To evaluate a proposed size, solve the network for representative data patterns and process corners, then include sense noise, timing, supply and reference uncertainty. A single two-point NL cannot supply these missing conditions.

Horizontal layers and vertical pillars

3D integration: selected path and inter-layer leakage



Leakage depends on structure and bias.

Horizontal stacking repeats lateral crossbar planes. In a vertical structure, memory cells occur where a pillar electrode meets separately addressable stacked word-line layers. The selected path and isolation depend on the particular layout and line bias.

Interlayer leakage, alignment, parasitic capacitance, contacts, heat and yield can offset the apparent density gain. Different 3D architectures cannot be ranked without a stated circuit and manufacturing comparison.

Interpret an area calculation correctly

For an idealized planar crosspoint with pitch $2F$ in each direction, area per cell is $(2F)^2 = 4F^2$. If $F = 20$ nm, this is $1600\,\mathrm{nm}^2$.

With three perfect layers over the same projected footprint and one bit per cell, the ideal projected cell area per bit is $1600/3 \approx 533.3\,\mathrm{nm}^2$. This is an accounting estimate, not the area of a manufactured memory chip.

Peripheral decoders, sensing, routing, redundancy, layer isolation, spacing and yield must be included in a system comparison. Extra layers can change process cost and thermal constraints.

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Self-selecting memory: the read path is state-dependent

A self-selecting cell combines nonvolatile storage and a sufficiently nonlinear access response in the same two-terminal stack. A separate selector is not required **only when** the complete array meets its leakage, disturb and sensing specifications at the chosen size and biases. It is not an ideal off-switch or a guarantee of bipolar operation.

Characterize both HRS and LRS, both polarities and pulse histories. Compare selected-state currents at V_{read} and the corresponding half-selected currents at the actual device voltages. A single NL value from one state cannot specify the full network or its retention and endurance.

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A conditional leakage budget

Assume a sense column has an ideal 0 V sense node, one selected cell at V_{read}, and 99 other branches each held in the LRS at the same half-select voltage. Each LRS branch conducts 10 nA at half bias and $10\,\mu$A at full bias, giving $NL = 1000$ for this state. The selected HRS current at full bias is $0.2\,\mu$A.

If the 99 branch currents remain fixed when the target changes state, their total is 99×10 nA $= 0.99\,\mu$A. The sensed HRS and LRS currents are 1.19 and $10.99\,\mu$A. The background exceeds the selected HRS current, but that alone does not imply a read error: evaluate the state distributions, sensing errors and threshold margin. A nonlinearity ratio alone does not establish the number, voltages or states of the leakage paths.

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Oxygen profiles are engineered across the stack

For a **particular VCM stack**, one design separates a vacancy-rich, relatively oxygen-poor region from a more nearly stoichiometric switching or tunnelling region. Layers of different composition and thickness can shape where oxygen vacancies are supplied and where a conductive path ruptures. The local profile is a design hypothesis to test, not a universal equilibrium diagram.

Reactive electrodes such as Ti or Hf may exchange oxygen with an adjacent oxide; an opposite, less reactive contact can contribute to an asymmetric boundary. Electrode oxidation, diffusion through imperfect barriers and ambient exposure can alter that intended profile. The electrochemical potentials, contact reactions and manufacturing history must be specified before predicting a direction.

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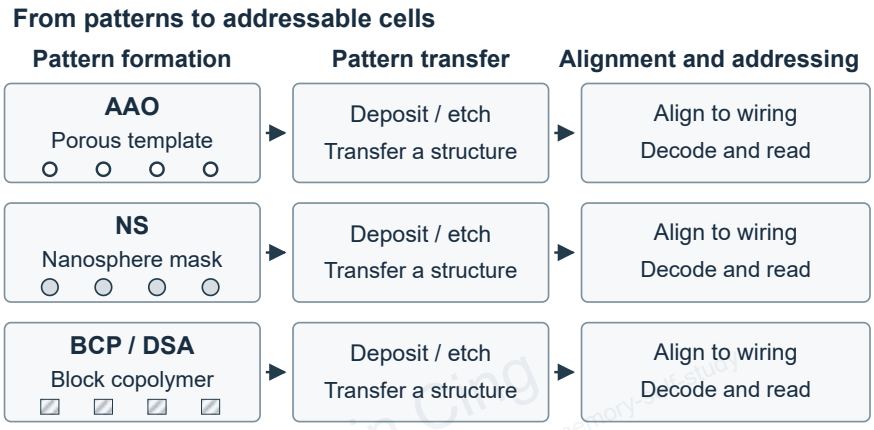
Dopants, alloys and buffer layers: what is controlled?

In VCM, dopants can alter vacancy formation energy, charge compensation and migration barriers. A material's response depends on dopant concentration, position, charge state and processing; the name of a dopant alone does not guarantee forming-free operation or improved retention.

In ECM, Cu or Ag from an active electrode supplies mobile ions. An alloy can change the chemical activity and release of the active species; a buffer layer can restrict unwanted diffusion or modify deposition geometry. Counter-electrode kinetics and electrolyte structure also affect the path. Test forming, switching variability, retention and disturbance together rather than optimizing only mobility.

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Three templates, one addressing problem



A regular pattern is not automatically addressable: contacts, alignment and defects matter.

AAO pores, nanosphere masks and guided block-copolymer domains form patterns by different processes. After transferring the pattern into the intended layer, registration to the electrode lines and continuity of each contact must be verified. A conductive-probe measurement at one site is useful evidence for that site; it is not by itself an electrically addressed memory array. A continuous switching layer may also create lateral leakage and crosstalk between cells.

Compare formation, transfer and registration

Method	Form a template	Transfer and registration
AAO	Anodize Al to generate pores; optionally open the pore bottoms.	Deposit into or through pores, or transfer the template. Guided initial patterning and alignment are separate engineering tasks.
Nanosphere lithography	Assemble a sphere monolayer.	Use spheres or their openings as a deposition/etch mask; remove the mask, inspect defects and align to contacts.
BCP / DSA	Anneal phase-separated polymer domains; a guide pattern constrains their location.	Selectively remove or convert one domain, transfer by etch or lift-off, then verify registration and isolation.

Template geometry, defect density, process compatibility and alignment depend on the actual process; these methods cannot be ranked from nominal feature size alone.

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Individual success versus a completely good array

Suppose 100 independently fabricated contacts each pass a specified electrical test with probability $p = 0.99$. With no repair or redundancy, the probability that **all** pass is

$$P(\text{all pass}) = 0.99^{100} \approx 0.366.$$

If X counts passing contacts, linearity of expectation gives $E[X] = 100(0.99) = 99$. This mean needs equal marginal pass probabilities but **not** independence; the all-pass product requires independence. Spatially correlated defects or repair change the whole-array yield, and a successful contact alone does not establish a functioning memory cell.

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A reproducible array comparison

State the stack, access structure, array dimensions, line biases, reference polarity, wire and sense models, pulses and timing. Record whether the reported current includes unselected paths and whether the result comes from measurements, simulation or extrapolation.

Report temperature, sample count, cycle history, variation, failure criterion and relevant uncertainty. A particular experimental selector or 3D stack is evidence for that configuration, not a universal specification for all RRAM arrays.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Redox mechanisms: from ion motion to memory

An oxide memory is a mixed ionic–electronic system. Its electrical response depends on mobile defects, electrode reactions, thermal history and measurement time. This unit develops the conditions needed to connect an observed I–V curve to a physical mechanism.

All numerical examples are hypothetical teaching models; they are not device specifications or reported measurements.

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Separate three observations

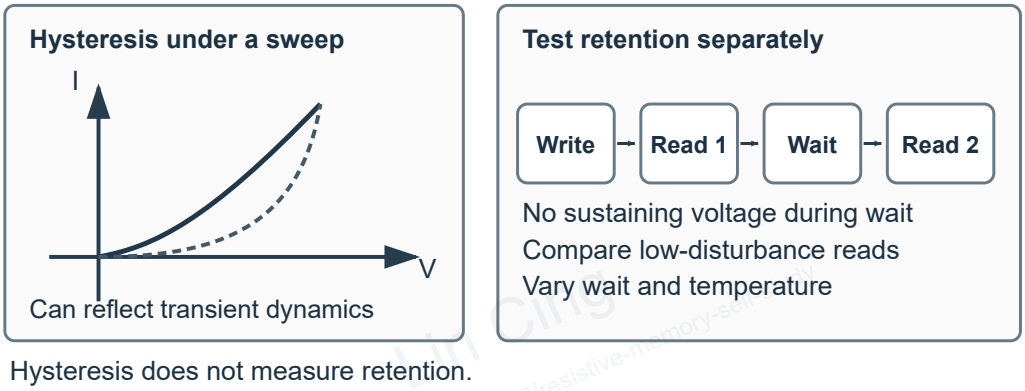
Switching is a change in conductance caused by an operation. **Retention** asks whether a distinguishable state remains after the driving signal is removed. **Short-term hysteresis** means the response depends on recent history over the measurement time.

A loop in a voltage sweep can occur when ion and electron responses have different timescales. It does not, by itself, establish long-term nonvolatile memory. Conversely, two retained states can be distinguished using a suitably non-disturbing read protocol without relying on the shape of a continuous sweep.

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A measurement distinguishes timescales

I–V hysteresis alone does not prove nonvolatile memory



Hysteresis does not measure retention.

The sketch compares two protocols; it does not assert a specific material or quantitative retention time. The waiting period must be specified and the read pulse checked for disturbance.

From material change to terminal current

One possible chain is: an applied pulse changes an electrode reaction and/or moves ionic defects; a nonuniform composition develops; that distribution modifies an electronic conduction path or interface barrier; the read current changes.

These links are hypotheses to be tested. An I–V loop alone cannot tell whether conduction is filamentary, interface-controlled, or distributed across the film. A mechanism should also predict spatial, chemical and time-dependent observations.

Charge carriers and their roles

The electronic current can be dominated by electrons or holes, while much slower ionic defects alter the device structure. The measured terminal current can include electronic, ionic and displacement contributions:

$$I_{\text{total}} = I_e + I_{\text{ion}} + I_{\text{disp}}.$$

In a one-dimensional series cross section, charge continuity gives the same total current at each position; individual contributions can vary with position and time as charge accumulates. An ammeter does not automatically measure ion flow.

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Reaction bookkeeping: an oxygen vacancy

For oxygen leaving an oxide into an oxygen reservoir, one possible idealized reaction is

$$O_O^{\times} \rightleftharpoons \tfrac{1}{2}O_2 + V_O^{\bullet\bullet} + 2e'.$$

Here $O_O^{\times}$ is oxygen on its lattice site, $V_O^{\bullet\bullet}$ is a vacancy with effective charge +2 relative to the lattice, and e' denotes an electron with effective charge −1. Both atom count and effective charge balance. This is a model of one charge state, not a claim that all vacancies have this charge.

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When may vacancy concentration predict electrons?

If vacancies are the only significant positively charged defects, electrons the only significant negative carriers, and local charge neutrality applies, then $n \simeq 2[V_O^{\bullet\bullet}]$.

For $[V_O^{\bullet\bullet}] = 2 \times 10^{18}\,\mathrm{cm}^{-3}$, the model gives $n \simeq 4 \times 10^{18}\,\mathrm{cm}^{-3}$. With negatively charged acceptors included, neutrality becomes $2[V_O^{\bullet\bullet}] = n + N_A^-$: for $[V_O^{\bullet\bullet}] = 3 \times 10^{18}$ and $N_A^- = 2 \times 10^{18}\,\mathrm{cm}^{-3}$, $n = 4 \times 10^{18}\,\mathrm{cm}^{-3}$ instead of $6 \times 10^{18}\,\mathrm{cm}^{-3}$ when acceptors are incorrectly omitted. Interface space charge can also invalidate local neutrality.

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ECM: metal transfer and reduction

In an electrochemical metallization (ECM) cell with an active Ag electrode, a possible oxidation reaction is $\mathrm{Ag} \rightarrow \mathrm{Ag}^+ + e^-$. Ag ions enter the switching layer; reduction $\mathrm{Ag}^+ + e^- \rightarrow \mathrm{Ag}$ can deposit metal elsewhere.

Where growth begins and how a path forms depend on ion transport, electron supply, interfaces and rates. The equations do not imply that every ECM cell has the same filament-growth direction or that SET has one universal polarity.

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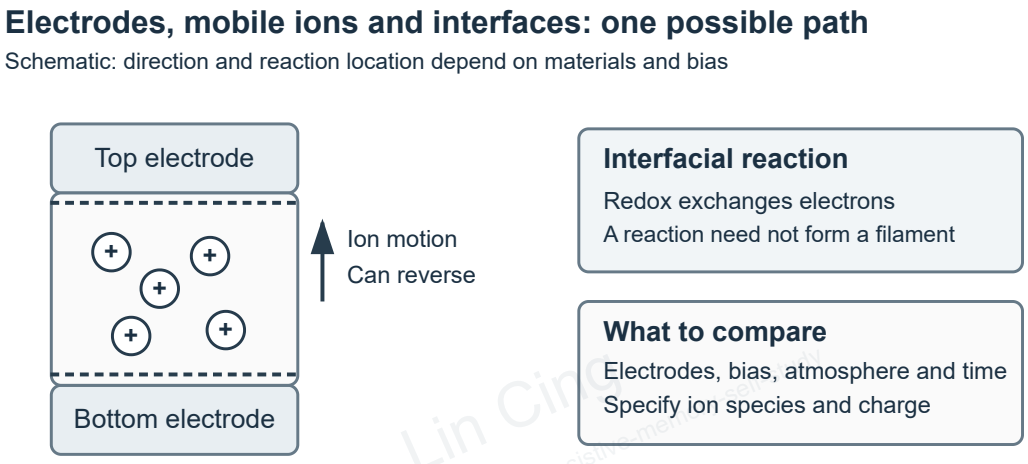
VCM: oxygen exchange and vacancy redistribution

In a valence-change mechanism (VCM), oxygen chemical potential, vacancy populations and local transition-metal valence can change. A reduced path, a narrow gap near an electrode, or a modified interface region can alter electronic transport.

VCM need not mean a metallic filament; ECM need not make every observed filament metallic throughout its lifetime. Composition-sensitive and spatially resolved measurements are needed to distinguish models.

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Follow ions and interfaces in a cell



The illustration shows a possible direction of ion motion, not an assigned electrode polarity or a specific reaction. For an Ag-containing ECM stack, the reactions on the preceding page are examples; actual growth direction depends on the stack, voltage and kinetics.

Electrochemical potential and driving force

For species i with charge number z_i, electric potential ϕ and chemical potential μ_i, the electrochemical potential is

$$\widetilde{\mu}_i = \mu_i + z_i e \phi.$$

At fixed temperature, a gradient of $\widetilde{\mu}_i$ drives transport. A concentration gradient contributes through μ_i; the electric field $E = -\partial\phi/\partial x$ contributes through $z_i e\phi$. Electric field alone does not determine net transport when concentration gradients are present.

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Drift and diffusion in a limited model

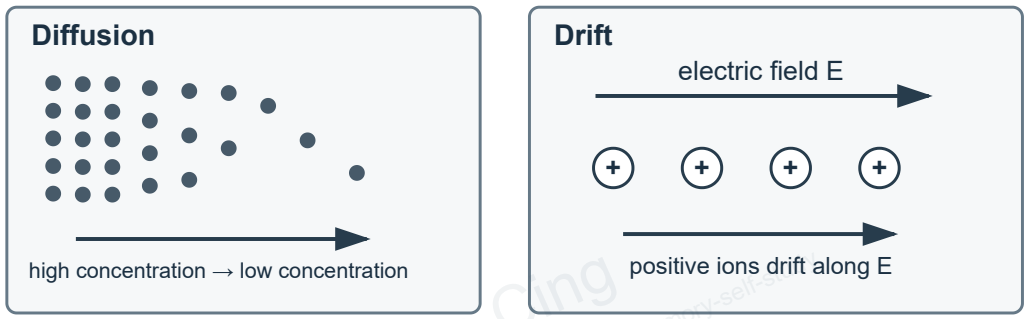
For dilute, noninteracting particles with concentration $c(x)$, constant D, charge number z, and the Einstein relation, the one-dimensional number flux is

$$J = -D\frac{\partial c}{\partial x} + \frac{Dze}{k_B T} cE.$$

Here J is number per area per time, not electrical current density; the latter is $j_{\mathrm{ion}} = zeJ$. For concentrated defects, strong fields, interactions or nonuniform temperature, use an appropriate nonideal transport model rather than this expression unchanged.

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Compare diffusion with electric drift



Net flux combines both terms and boundary conditions; electron flow is separate.

The arrows are for the stated positive defect and illustrated gradients only. Changing charge sign or either gradient can reverse an individual contribution; the net flux is their vector sum.

Drift and diffusion: calculate both terms

For the dilute model, let $D = 10^{-18}\,\mathrm{m^2/s}$, $z = 2$, $E = 10^{8}\,\mathrm{V/m}$, $c = 10^{24}\,\mathrm{m^{-3}}$, $dc/dx = 10^{30}\,\mathrm{m^{-4}}$ and $k_B T = 4.14 \times 10^{-21}$ J.

The diffusion flux is $-10^{12}\,\mathrm{m^{-2}s^{-1}}$. With $e = 1.60 \times 10^{-19}$ C the drift contribution is $+7.73 \times 10^{15}\,\mathrm{m^{-2}s^{-1}}$; at this precision the net flux is also approximately $+7.73 \times 10^{15}\,\mathrm{m^{-2}s^{-1}}$. These are particle fluxes, not amperes.

A field does not erase a migration barrier

Suppose a 12 nm film has a uniform 0.6 V drop. The electric-field magnitude is 0.05 V/nm. Across a 0.2 nm hop along the electric field the potential falls by 0.01 V; for a hypothetical $z = +2$ defect the endpoint electrostatic energy changes by -0.02 eV.

This compares endpoint energies. The reduction of the actual activation barrier depends on the transition-state position and energy landscape and is not automatically 0.02 eV. A highly nonuniform drop at an interface changes the estimate.

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Interface reaction versus bulk transport

An electrode can exchange oxygen with the oxide, store oxygen, react to form an interlayer, or approximately block exchange. A blocking boundary constrains normal flux there; it does not imply that the defect concentration is uniform inside the film.

If supply at one interface is slow compared with redistribution in the oxide, defects can accumulate in a different region from the fast-supply case. Growth direction and the active gap therefore depend on **both** boundary reaction kinetics and bulk transport, not electrode name alone.

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A supply–transport balance, with limits

For one illustrative slab of thickness L and cross-sectional area A, let $N(t)$ be its spatially averaged defect concentration, with injection flux s and removal flux r on opposite boundaries. With no net volumetric reaction, conservation gives $d(ANL)/dt = A(s - r)$, hence

$$\frac{dN}{dt} = \frac{s - r}{L}.$$

Fluxes may depend on concentration, voltage, temperature and time. This averaged balance cannot determine a local profile, filament geometry, or which side the filament grows from; those require the transport equation and boundary conditions.

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Crossing an assumed concentration threshold

Take the simplified constant-flux model with $L = 10\ \text{nm} = 10^{-6}$ cm, $s = 8 \times 10^{12}$ and $r = 2 \times 10^{12}\ \text{cm}^{-2}\text{s}^{-1}$. Then $dN/dt = 6 \times 10^{18}\ \text{cm}^{-3}\text{s}^{-1}$.

If $N(0) = 2 \times 10^{18}\ \text{cm}^{-3}$ and a specified model threshold is $N_c = 5 \times 10^{18}\ \text{cm}^{-3}$, then $N(t) = N(0) + (s - r)t/L$ reaches N_c after $(5 - 2)/6 = 0.5$ s. This is an average crossing, not proof of a local metal transition; a real threshold depends on material and spatial distribution.

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A separate steady-state spatial model

Now consider a different one-dimensional model: constant D, no drift, no volume sources, and fixed endpoint concentrations. Steady conservation gives $dJ/dx = 0$, and Fick's law gives $J = -D\,dc/dx$. Thus $c(x)$ is linear and the flux is uniform.

Take $L = 10\ \text{nm} = 10^{-6}$ cm, $D = 10^{-14}\,\text{cm}^2/\text{s}$, $c(0) = 8 \times 10^{18}$ and $c(L) = 2 \times 10^{18}\,\text{cm}^{-3}$. Then

$$c(x) = \left(8 - 6\frac{x}{L}\right) 10^{18}\,\text{cm}^{-3}, \quad J = +6 \times 10^{10}\,\text{cm}^{-2}\text{s}^{-1}.$$

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Where does the local threshold hold?

Using that steady diffusion solution and the illustrative threshold $N_c = 5 \times 10^{18}\,\text{cm}^{-3}$, solve $8 - 6x/L \geq 5$: the interval is $0 \leq x \leq L/2$, or 0–5 nm, of length 5 nm.

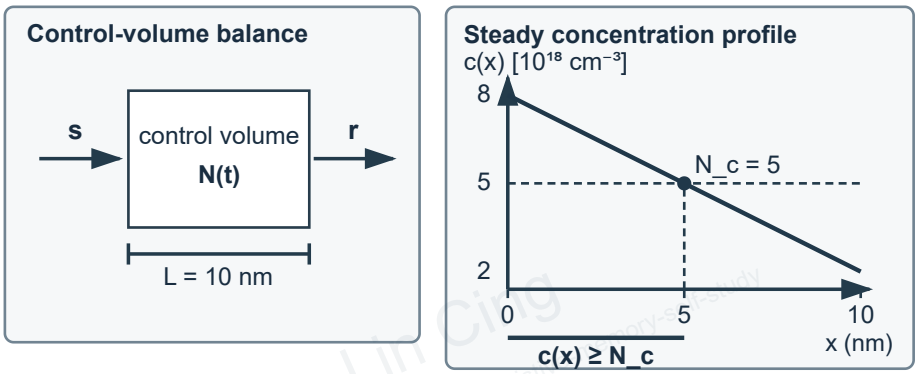
The assumed boundary concentrations and steady-state profile differ from the earlier transient control-volume example; do not combine the two as one time history. Crossing a chosen concentration threshold in this 1D model is not a general prediction of a connected, metallic filament.

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Boundary flux and local profile

Control-volume balance and a concentration profile

Teaching examples: the two models are separate



A threshold alone proves neither phase change nor filament.

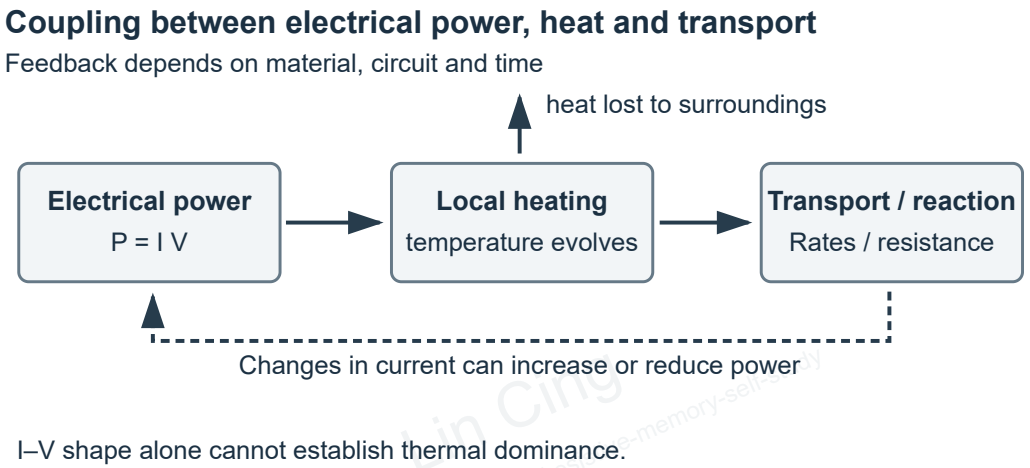
The left panel is a transient average; the right panel is a separate fixed-boundary steady state. Their fluxes and concentrations are not one continuous experimental sequence.

Forming, compliance and feedback

Some devices require electroforming: an initial electrical treatment that establishes a switchable state. Field-driven reactions and local Joule heating can reinforce one another; the evolving structure changes the local field and current.

Compliance limits the terminal current under the stated circuit and timing. Finite response time and parasitic capacitance can still produce a transient overshoot, so the set point alone does not fix local peak current, temperature or filament dimensions. Other devices are forming-free under their specified preparation and test conditions.

Feedback can amplify a local change



I–V shape alone cannot establish thermal dominance.

This loop is a possible mechanism under a specified circuit: increasing temperature need not increase conductance in every material or bias regime. Compliance and thermal loss limit the process differently.

Why a nanometre gap matters

Take a deliberately simple model in which read current through a barrier is $I(g) = I_0 e^{-2\kappa g}$ with fixed I_0 and $\kappa = 5\,\mathrm{nm}^{-1}$. A gap change from $g_1 = 1.0$ nm to $g_2 = 1.3$ nm gives

$$\frac{I(g_2)}{I(g_1)} = e^{-2(5)(0.3)} = e^{-3} \simeq 0.0498.$$

Thus the wider-gap current is about 5% of the initial current. This is an exponential sensitivity illustration, not a complete tunnelling theory: barrier height, bias, geometry and prefactor may also change.

Interface control is not proof of a filament

An interface barrier can limit the current even when the remainder of the oxide is relatively conductive. Changes in charge distribution and local doping can modify barrier width, band bending and available transport channels.

At fixed temperature and device geometry, ask whether an area series, local current map, spectroscopy and cross-sectional imaging support a distributed interface or localized path. No single area-scaling result proves the entire current topology without checking confounders.

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Bipolar and unipolar are operating descriptions

In **bipolar** operation, the two transitions are induced under opposite voltage polarities as defined for that device. In **unipolar** operation, both can occur at the same polarity under different pulse, current, thermal or circuit conditions.

The label classifies a protocol, not a unique microscopic mechanism. In particular, same-polarity RESET does not prove a particular temperature or diffusion process without supporting measurement.

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Compare the energy of voltage scans

For a voltage ramp beginning at zero with constant positive scan rate $v = dV/dt$, substitute $dt = dV/v$ into $W = \int V I \, dt$ to obtain

$$W = \frac{1}{v} \int_0^{V_{\mathrm{end}}} V I(V) \, dV.$$

Use the current actually measured along each scan, including its state history; this change of variables does not assume SET and RESET share the same I–V curve. Different ramp rates and switching points must be included explicitly.

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A same-polarity scan comparison

Take two hypothetical positive-going scans, each at $v = 1$ V/s. For the SET scan let $I = aV$ with $a = 1\,\mu$S up to 1 V; for RESET use $a = 4\,\mu$S up to 0.5 V. Integrating $VI = aV^2$ gives

$$W_{\mathrm{SET}} = \frac{1\,\mu\mathrm{S}}{1\,\mathrm{V/s}} \frac{1^3}{3} = \tfrac{1}{3}\,\mu\mathrm{J}, \quad W_{\mathrm{RESET}} = \frac{4\,\mu\mathrm{S}}{1\,\mathrm{V/s}} \frac{0.5^3}{3} = \tfrac{1}{6}\,\mu\mathrm{J}.$$

Higher current at a given voltage alone does not determine integrated energy when voltage ranges and durations differ. Even if one scan delivers more energy, that alone cannot determine which reaches a higher peak temperature: the heat-loss path and time dependence also matter.

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Power, energy and temperature are different

Instantaneous electrical power is $P(t) = V(t)I(t)$ and delivered energy is $W = \int P(t)dt$. The same energy can produce different peak temperatures if pulse widths, heat loss and thermal paths differ.

For a lumped model, $C_{\mathrm{th}}\, d\Delta T/dt = P(t) - \Delta T/R_{\mathrm{th}}$. With constant power, $\Delta T(t) = PR_{\mathrm{th}}(1 - e^{-t/(R_{\mathrm{th}}C_{\mathrm{th}})})$. This model is not a spatial temperature map.

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A thermal pulse worked through

Assume $V = 1$ V and $I = 100\,\mu$A remain constant, so $P = 0.1$ mW. Let $R_{\mathrm{th}} = 10^6$ K/W, $\tau_{\mathrm{th}} = 100$ ns, and initially $\Delta T = 0$. The implied $C_{\mathrm{th}} = \tau_{\mathrm{th}}/R_{\mathrm{th}} = 10^{-13}$ J/K; the steady rise is $PR_{\mathrm{th}} = 100$ K.

After 100 ns the idealized rise is $100(1 - e^{-1}) \simeq 63.2$ K. Changing the thermal resistance, heat capacity or pulse shape changes the result; measuring electrical energy alone does not reveal peak local temperature.

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Retention and activation: a toy estimate

An activated process is often approximated by $\tau = \tau_0 \exp(E_a / k_B T)$ when one rate-limiting barrier dominates and temperature is constant. For $\tau_0 = 10^{-12}$ s, $E_a = 1.0$ eV and $k_B T = 0.02585$ eV at 300 K,

$$\tau \simeq 10^{-12} e^{38.68} \simeq 6.3 \times 10^{4}\ \mathrm{s} \approx 17.5\ \mathrm{h}.$$

This hypothetical time is not a retention prediction for a particular device. Real retention depends on barrier distributions, geometry, read criterion and other mechanisms; the fitted prefactor must be stated.

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Read–wait–read: test memory directly

Program two populations to HRS and LRS, remove the programming bias, wait at controlled temperature, then apply the same low-disturb read to each. Repeat for multiple times and devices; report read voltage, pulse duration, temperature and the state distributions.

Retention is a criterion on the separation and error rate of those distributions, not merely a single resistance value. Check that repeated reads do not themselves change the state; include drift and uncertainty.

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What an I–V crossing can and cannot show

In a simplified passive, non-battery-like device model, negligible displacement and ionic currents at low frequency can make the terminal curve pass near the origin. Even if a curve passes exactly through the origin, that is **not sufficient** to establish retention; an internal state may relax.

Offsets from capacitance, electrochemical electromotive forces, instrumentation and finite sweep rate can shift the crossing. These assumptions must be checked before interpreting a crossing. Compare sweep rates and perform the read–wait–read experiment.

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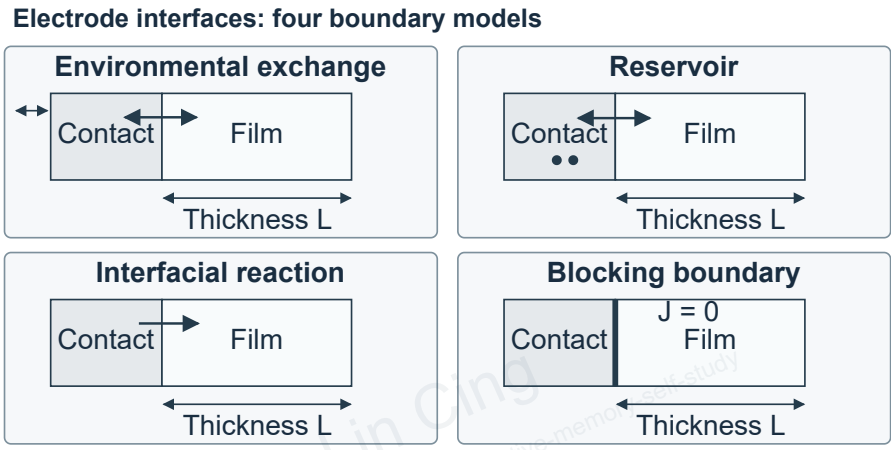
Build a falsifiable mechanism

1. Specify electrode stack, polarity convention, forming and compliance circuit, pulse timing, atmosphere and temperature.
2. Compare hypotheses using predictions for spatial current, element distribution, state evolution and temperature—not only a fit to one I–V curve.
3. Test alternative explanations and controls, including interface changes, parasitic current, measurement history and sample-to-sample variation.

An original model is useful when it explains the data within its assumptions and can be challenged by another experiment.

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Electrode boundaries determine the oxygen inventory



Boundary depends on material and bias; arrows show possible flux.

An electrode may allow exchange with the environment, store oxygen in grain boundaries, react to form a reservoir layer, or approximately block exchange. The substrate, edge leakage and actual interface matter: a nominally inert electrode is not necessarily perfectly blocking. These labels describe a particular stack and timescale, not permanent material constants.

Count oxygen and vacancies at the boundaries

For an oxide slab of thickness L, take vacancy number flux $J(0)$ as entering at $x = 0$ and $J(L)$ as leaving at $x = L$. With no volume generation or reaction, integrating $\partial c/\partial t = -\partial J/\partial x$ gives

$$\frac{d\bar{c}}{dt} = \frac{J(0) - J(L)}{L}.$$

In a closed, blocking stack, both boundary fluxes vanish and the total vacancy inventory is conserved, although its *spatial distribution* can change. An interfacial reservoir or a reaction adds its own inventory and kinetics; include these in the total balance rather than assuming a fixed oxide inventory. Oxygen and vacancy fluxes require a consistent sign convention and are not interchangeable without a defect model.

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Quantify net boundary exchange

For example, a net inward vacancy flux $J(0) - J(L) = 10^{12}\,\mathrm{cm^{-2}s^{-1}}$ across $L = 10$ nm raises the average concentration by $2 \times 10^{18}\,\mathrm{cm^{-3}}$ in 2 s. A possible linear exchange boundary at $x = 0$ is $J(0) = k[c_{\mathrm{eq}} - c(0)]$, with k in cm/s; its parameters and the opposite boundary must be specified to solve the spatial profile.

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Supply versus transport: compare two limiting pictures

Take the book's particular forming convention: cathode at $x = 0$ on an oxygen-blocking substrate, anode at $x = L$. If vacancy supply at the anode is limited while bulk removal is relatively fast, a high-vacancy region can accumulate toward the blocking cathode and its boundary may move toward the anode. If anode supply remains relatively fast compared with bulk removal, a high-concentration region can instead extend from the anode while a low-concentration gap remains near the cathode.

These are conditional qualitative pictures, not a formula that maps polarity to a unique front. Rate relationships may reverse as conductivity and fields change, and the assumed threshold may never be reached. Electrode geometry, reactions and the actual field profile must be measured or modelled.

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Forming and RESET are different boundary problems

In one bulk-reduction scenario, forming increases vacancies without requiring the whole oxide to cross the assumed metallic threshold. During a subsequent RESET, the former anode can become the cathode; fast vacancy removal at that boundary can leave a local low-concentration, high-resistance gap. A later SET may replenish it under the stated reactions and boundary conditions.

This is one proposed explanation, not a prediction for every device. Reversing the terminal voltage changes electrode roles only under the defined voltage convention; a Schottky barrier, interfacial reactions, heat or another transport pathway can instead dominate the observed switching.

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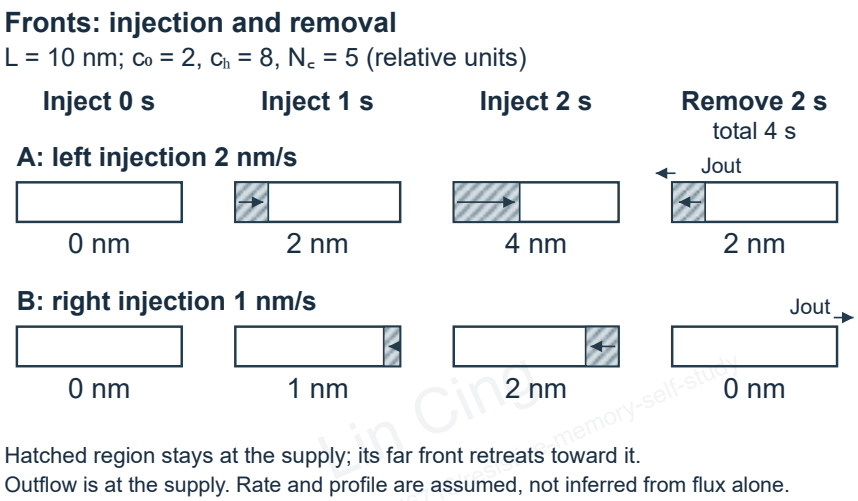
A conservation-based moving-front example

For a separate illustrative sharp-front model, let $L = 10$ nm, $c_{\mathrm{low}} = 2$ and $c_{\mathrm{high}} = 8$ in units of $10^{18}\,\mathrm{cm}^{-3}$, and let the chosen threshold be $c_{*} = 5$ in the same units. In case A, the high region starts at $x = 0$ and grows at 2 nm/s; in case B it starts at $x = L$ and grows at 1 nm/s. At $t = 2$ s, the above-threshold lengths are 4 and 2 nm, respectively.

Before the front reaches a boundary, conservation requires net influx per area $s = (c_{\mathrm{high}} - c_{\mathrm{low}})v$: 1.2×10^{12} and $6 \times 10^{11}\,\mathrm{cm}^{-2}\mathrm{s}^{-1}$. The model prescribes both the front and its speed; it does not derive the moving front from the book's microscopic kinetics.

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Follow the two prescribed fronts



Hatched region stays at the supply; its far front retreats toward it.
Outflow is at the supply. Rate and profile are assumed, not inferred from flux alone.

The two configurations have separate boundary conditions. Arrows and front speeds are assumptions of this example, not a universal relationship between growth direction and terminal polarity.

Track the RESET front without inventing a mechanism

Now assume each high-concentration region remains anchored at its original supply interface, and its far front recedes toward that interface at 1 nm/s for 2 s. Assume internal redistribution maintains this shape and the corresponding net loss leaves through that interface. The lengths change A: $4 \to 2$ nm and B: $2 \to 0$ nm; A retains $[0, 2]$ nm. The magnitude of the net outward vacancy flux is $(8 - 2) \times 10^{18} \times 10^{-7} = 6 \times 10^{11}\,\mathrm{cm^{-2}s^{-1}}$ while the front exists.

This geometry and redistribution are deliberately assigned; the net flux alone does not uniquely determine either. This toy history is separate from the qualitative source-limited and transport-limited pictures above. An above-threshold interval in one dimension alone does not prove a three-dimensional connected filament, a metallic phase, or a unique SET/RESET polarity.

References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Seeing a switching device: measurements and evidence

In a nanoscale device, electrical response is a measured outcome; a filament, its composition and its motion are explanations that require additional evidence. This chapter connects an experiment, its signal and the limits of the conclusion.

Numerical examples are hypothetical. Images are original schematics; experimental figures and published performance values are not reproduced.

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Start with an evidence chain

Ask four questions in order: **What was driven? What was measured? How was it calibrated? Which mechanisms remain compatible with the data?**

An I–V change shows an electrical state change under the stated protocol. A time-correlated image supplies structural evidence; chemical analysis and control measurements can narrow the interpretation. No single contrast feature automatically establishes composition, valence, or a unique atomic mechanism.

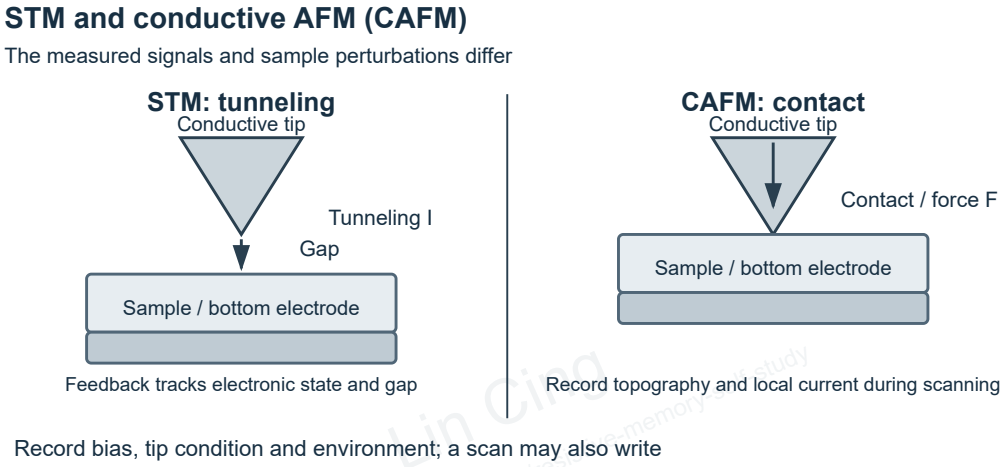
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Choose the probe for the question

Method	Primary signal	Useful question	Main limitation
in situ TEM	Transmitted-electron image, optionally EDS/EELS	How does a visible structure evolve under bias?	Thin specimen, beam and vacuum may perturb it.
STM/STS	Tunneling current or differential conductance	How does a local surface electronic response change?	Surface, tip and feedback influence the signal.
Conductive AFM	Contact current and topography	Where does current enter an accessible surface?	Contact and scan can alter the sample.
Scalpel SPM / PMCM	Sequential contact maps / force-modulated electrical response	What depth structure or mechanically sensitive region is inferred?	Reconstruction or pressure introduces additional assumptions.

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Link the device and the instrument



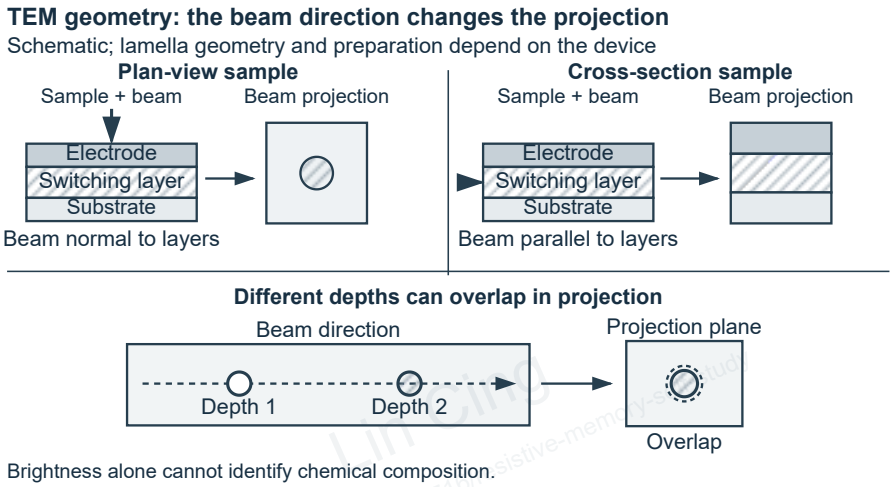
Record geometry, electrode assignments, voltage polarity, compliance, timing, environment and prehistory. The probe's electrical boundary is part of the experiment: a tip electrode is not automatically equivalent to an intact device's top electrode.

In situ TEM: what is actually observed?

An electron beam passes through a sufficiently thin specimen. The image is a **projection** of electron interactions through its thickness, not a direct three-dimensional concentration map. Biasing the specimen while acquiring images allows the electrical trace and structural evolution to be compared in time.

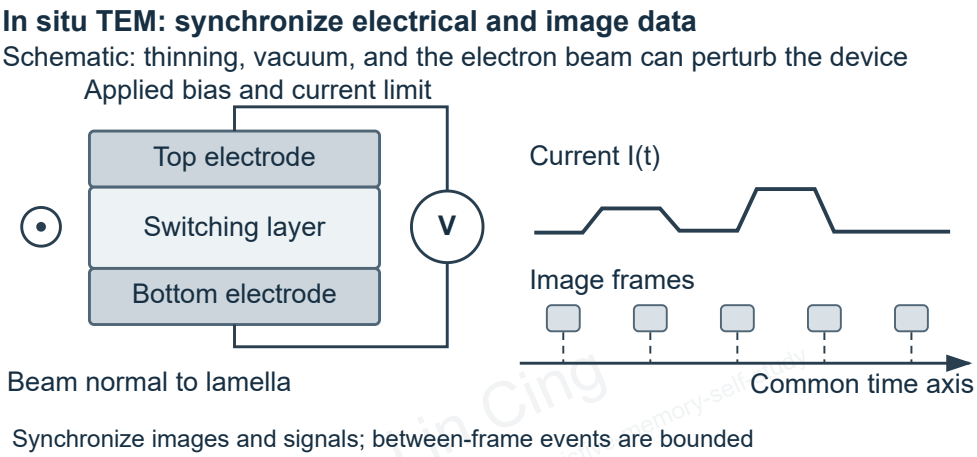
Different growth morphologies and nucleation positions are possible in different materials. A particular device's polarity or shape does not fix the direction for every ECM cell.

Cross-section and plan view: projections can overlap



In a cross-section view, objects separated along the beam can overlap as one feature in projection. For a deliberately ideal linear projection, two contributions of 2 and 3 arbitrary signal units sum to 5; that observed value alone cannot identify how many objects produced it. Real TEM contrast may be nonlinear and is not generally an additive concentration measurement. A plan-view projection looks from another orientation but still integrates along its own beam direction; tilting and matched electrical/chemical evidence help resolve the ambiguity.

Synchronize image and electrical trace



Before interpreting a feature, document which event is aligned with each frame, the exposure and clock uncertainty, and whether the device still switches after thinning. A frame is not necessarily an instantaneous sample: an exposure can average changes occurring during its interval.

A time-bound example

Suppose ideal instantaneous images are acquired every 30 ms. Frame 1 occurs at $t = 0$; frame 3 at 60 ms has no visible feature and frame 4 at 90 ms has one.

Under the ideal assumption that the feature becomes detectable once and remains detectable, its first detectable appearance is in $(60, 90]$ ms. The sequence alone does **not** give its exact onset, a speed or a chemical identity. Finite exposure, synchronization error and detection limits widen or complicate this inference.

Contrast, composition and oxidation state

Image contrast depends on specimen geometry, thickness, crystal orientation and imaging conditions as well as composition. A bright or dark region is evidence of contrast, not by itself a measurement of Ag, Cu or oxygen-vacancy concentration.

EDS provides elemental spectral information; EELS can provide elemental and chemical-state information when energy resolution, background and reference spectra support it. Both average over a finite interaction volume and can change the dose applied to the sample.

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TEM controls make a mechanism more credible

Compare an unbiased or lower-dose control, multiple switching cycles, and electrical behavior before and after imaging. Examine whether a morphology change coincides with a state change and whether composition-sensitive measurements support the proposed material.

FIB preparation, exposed surfaces, vacuum and the beam can affect oxygen content, interfaces, heating or damage. A non-switching thinned specimen cannot be silently treated as an undisturbed working device.

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Apparent motion has more than one explanation

Consider a cluster seen at position A in one frame and position B in a later frame. Translation is one hypothesis; dissolution near A and nucleation near B is another. Motion between frames or below resolution is not directly observed.

Vary time resolution and operating conditions, compare repeated trajectories and seek chemical or structural evidence. Report a mechanism as a supported model, not as the only possible explanation, when alternatives remain.

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STM: current across a tunneling gap

A conducting tip approaches a surface without the normal mechanical contact used in contact-mode AFM. At a fixed bias and small gap, the tunneling signal is strongly gap-dependent. Local surface states, tip states, the bias convention and the environment also influence current.

A pristine insulating film may not provide a stable measurable tunneling path; interpreting an image requires a specified sample stack and accessible surface.

In a simplified fixed-barrier model, $I \propto e^{-2\kappa z}$. If $\kappa = 10\,\mathrm{nm}^{-1}$ and the gap increases by 0.10 nm with all other parameters held fixed, $I_2/I_1 = e^{-2} \approx 0.135$. This is a model example, not a universal STM calibration.

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Feedback changes the meaning of an image

In **constant-current STM**, a feedback loop adjusts tip height to maintain a current setpoint while scanning. The recorded height reflects both geometry and local electronic response; a higher trace is not necessarily a physical protrusion.

In spectroscopy, an I–V trace and dI/dV depend on position, tip, bias, temperature and measurement protocol. Differential conductance may inform electronic states under a stated model; it is not an uncalibrated elemental map.

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A local write-and-read control

1. Image a region at a read condition chosen to minimize disturbance, and choose an unprogrammed control region.
2. Apply a specified local write pulse; record tip, pulse polarity, amplitude and duration.
3. Repeat low-disturb read at the same positions, then read again after a specified wait.
4. Compare current, topography and control drift. A persistent difference supports a retained local change under this protocol; it does not alone identify its chemical origin.

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Conductive AFM: contact current and topography

CAFM uses a conducting tip in mechanical contact with an accessible surface. A bias applied between the tip and another electrode yields a local current while the feedback system records topography. Tip force, coating, contact area, humidity, bias and current limit all affect the result.

The current map shows the response of **tip + contact + exposed sample + external circuit**, not an isolated, geometry-free property of the oxide.

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Two CAFM experiments with different boundaries

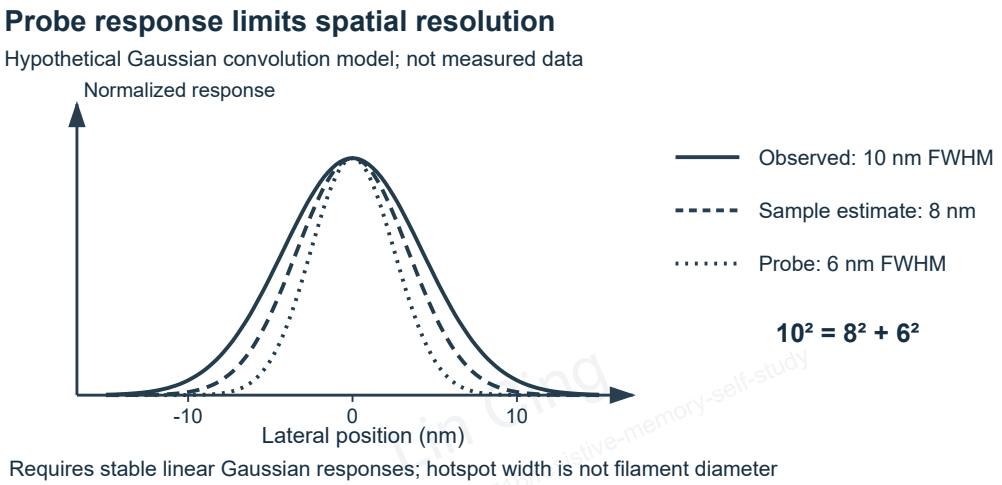
Tip-as-electrode: write and read directly on exposed film. The contact's field and force are part of the writing process.

Program first, expose later: switch a complete device electrically, remove its top electrode and map the exposed layer. Removal can damage or oxidize the feature, and the device cannot necessarily be cycled in the original configuration afterward.

Do not pool the two as if they had identical contacts, fields and sample histories.

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A current hotspot is not a direct diameter measurement



Measured width depends on the true spatial response, probe/contact geometry, sampling, feedback and noise. A visually wider hotspot can result from either a wider conducting region or a broader instrument response. Fit a stated forward model and show its limitations before inferring a size.

Gaussian resolution model: work through the numbers

Assume the true one-dimensional signal and a linear, shift-invariant point-spread function are Gaussian, with no additional broadening. Gaussian variances add; because $\mathrm{FWHM} = 2\sqrt{2\ln 2}\,\sigma$,

$$w_{\mathrm{obs}}^2 = w_{\mathrm{sample}}^2 + w_{\mathrm{tip}}^2.$$

For $w_{\mathrm{obs}} = 10$ nm and $w_{\mathrm{tip}} = 6$ nm, $w_{\mathrm{sample}} = \sqrt{10^2 - 6^2} = 8$ nm. The result is a model-dependent effective width, **not** a universal CAFM diameter correction.

If the two FWHMs have independent standard uncertainties of 0.5 nm, first-order uncertainty propagation gives $\sigma_w = \sqrt{(10 \cdot 0.5)^2 + (6 \cdot 0.5)^2}/8 \approx 0.73$ nm. This does not include uncertainty from model mismatch.

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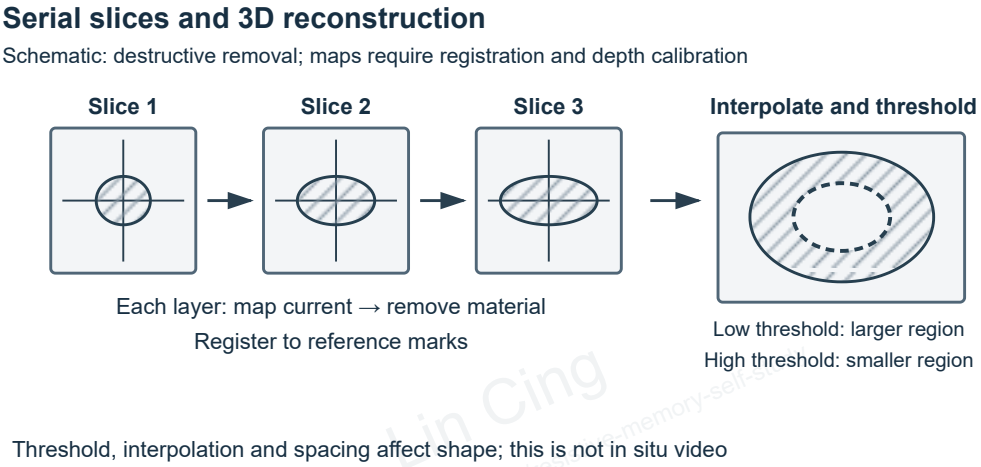
When deconvolution is not justified

If the measured width is no greater than the independently calibrated instrumental width, this Gaussian subtraction yields no positive real width. Within measurement uncertainty the feature may be unresolved; report an upper bound or a measurement limit rather than a fabricated diameter.

Even when subtraction is real, nonlinear contact, changing tip shape, non-Gaussian profiles and unknown uncertainty invalidate or weaken the estimate. Inspect raw profiles and calibration data.

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Scalpel SPM: a sequence, not a live 3D view



Program and locate a structure, remove material in controlled increments, and acquire a current map at each exposed surface. Register slices to a common reference and calibrate depth. The measurement is destructive and sequential; the sample may evolve between slices.

Reconstruction introduces choices

Spatial registration, depth estimates, interpolation, smoothing and the threshold used to draw an isosurface all affect the displayed shape. A continuous rendering is a visualization of measured slices **plus assumptions**, not independently measured data at every voxel.

Vary registration and threshold, use a stable reference electrode or separately characterized reference region at each slice to monitor tip changes, and report the slice spacing and removal uncertainty. A changing reference response flags possible tip wear or contamination; it cannot by itself rule out all local contact changes. Samples may oxidize or be mechanically damaged during exposure.

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Interpolate explicitly

At one registered lateral position, suppose measured currents are 2, 6 and 10 nA at depths $z = 0, 2, 4$ nm. With linear interpolation between the first two measurements,

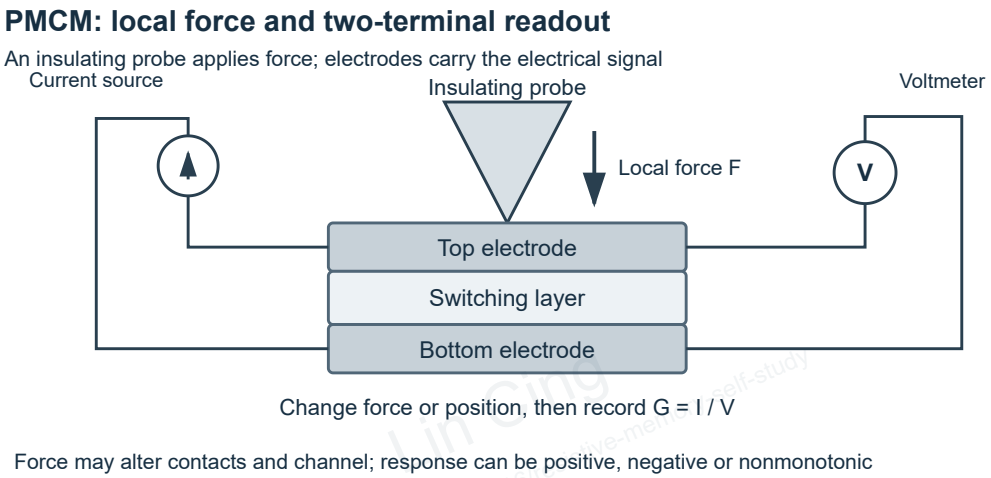
$$I(1\,\mathrm{nm}) = 2 + \frac{1-0}{2-0}(6-2) = 4\,\mathrm{nA}.$$

This 4 nA value is a model estimate, not a fourth measurement. An isosurface threshold of 3 nA includes different locations from one at 7 nA; a changing displayed volume is not by itself proof that the physical channel changed.

Similarly, interpolation between $z = 2$ and 4 nm gives $I(3) = 8$ nA and places the 7 nA crossing at $z = 2.5$ nm; both depend on the linear assumption.

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PMCM: a mechanical perturbation, an electrical readout



A non-conducting tip applies localized mechanical force to the top electrode of a complete device while the voltage across it is monitored under a low fixed current. Scanning the tip can reveal positions at which the whole-device resistance responds to that perturbation.

Force magnitude, load distribution, electrode mechanics, electrical bias, tip wear and spatial registration must be reported. The readout is not simply the current through the tip.

Calculate the conductance response

At fixed $I = 1\,\mu$A, suppose conductance rises from $G_0 = 10\,\mu$S to $G_1 = 12\,\mu$S when pressure is applied. Then

$$R_0 = 100\,\mathrm{k\Omega}, \quad R_1 = 83.33\,\mathrm{k\Omega}, \quad V_0 = 0.100\,\mathrm{V}, \quad V_1 \approx 0.0833\,\mathrm{V}.$$

The fractional conductance change is $(G_1 - G_0)/G_0 = 0.20$, or $+20\%$. A voltage dip at fixed current corresponds to lower resistance in this example; response sign and spatial shape depend on the device and mechanism.

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What a PMCM contrast does and does not show

A repeatable force-correlated response identifies a mechanically sensitive region under specified conditions. It can guide cross-sectional observation, but does not by itself establish that every current path is there, reveal a unique filament geometry, or prove that conductance must rise under pressure.

Tunneling distances, electronic structure, mechanical deformation and quantum interference can contribute differently. Compare reverse scans, force-dependence and controls with other imaging and electrical measurements.

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Coherent transport: why the response may change sign

For a phase-coherent conductor at zero temperature and infinitesimal bias, the Landauer expression is $G = G_Q \sum_n T_n(E_F)$, where $G_Q = 2e^2/h$ assumes spin degeneracy and T_n is the transmission probability of mode n. At finite temperature the energy-dependent transmission is weighted over the thermal window; finite bias requires an appropriate non-linear transport treatment.

For one illustrative lossless symmetric double barrier with single-barrier probabilities $T_b = 2/3$ and $R_b = 1/3$, summing the coherent multiple-reflection amplitudes gives $T(\phi) = T_b^2/|1 - R_b e^{i\phi}|^2 = 1/[1 + 3\sin^2(\phi/2)]$. The phase ϕ includes propagation and reflection contributions. This describes a model, not a measured device.

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Two phase points, opposite pressure responses

Assume, only for this example, that compression changes phase by $+\pi/6$ without changing the barriers. At $\phi = \pi/2$, T changes from $2/5$ to $4/13$; hence $\Delta G/G = -3/13 \approx -23.1\%$. At $\phi = 3\pi/2$, the same phase increment changes T from $2/5$ to $4/7$, giving $\Delta G/G = 3/7 \approx +42.9\%$.

The assumed phase shift is not a universal pressure law. Inelastic/dephasing processes, thermal averaging and several channels with different phases can weaken or alter the oscillation; other pressure-driven changes in barriers also violate the fixed-barrier model. An observed sign alone does not identify a unique mechanism.

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From observation to defensible claim

For every claim, attach (1) the measurement and sample preparation, (2) spatial and temporal resolution, (3) operating protocol and controls, and (4) competing interpretations.

Example: **Observed:** low-bias current map changes at a location after a pulse. **Supported:** a local electrical response changed under that protocol. **Not yet established:** that the spot is an isolated metal filament of the map's measured width. Composition and geometry require additional, matched evidence.

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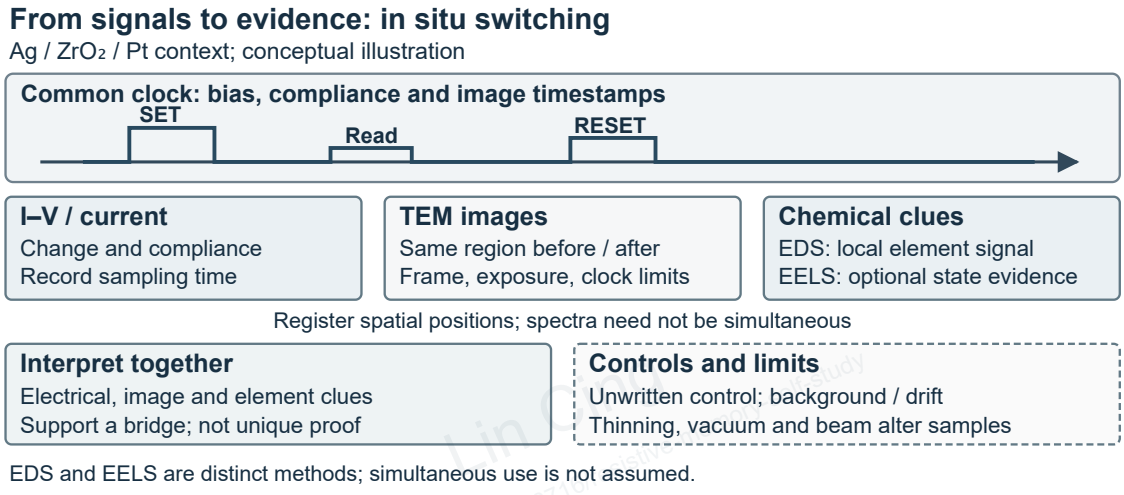
A practical experiment checklist

Stage	Questions to record
Before	Device stack and state; tip condition; alignment; environment; baseline trace.
During	Bias, compliance, current, force, acquisition time, clock synchronization and dose.
After	Repeatability, control regions, device survival, registration, uncertainty and alternative explanations.

Use the checklist to decide which follow-up experiment would distinguish two surviving mechanisms, rather than treating a visually persuasive image as a complete proof.

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A device-specific TEM evidence chain



In the reported FIB-prepared Ag/ZrO_2/Pt cell (approximately 100 nm scale), TEM images, SET/RESET I–V and EDS were compared. Ag-rich formation and dissolution were reported (source PDF p. 107). This figure is a general schematic, not the specimen or its data.

Align observations without inventing time resolution

For a proposed SET event, associate each image or chemical acquisition with its **actual exposure interval**, electrical trace and clock uncertainty. A post-SET EDS spectrum cannot be treated as a continuously recorded elemental movie. EELS is a complementary method in other experiments, not an assumed simultaneous measurement in this Ag example.

An EDS signal supports elemental attribution only with suitable local background and references; image contrast alone does not. Preparation, beam and vacuum may perturb the specimen: compare switching before and after imaging and use appropriate controls.

In a separate Pt/MoO_x/TiN video example, the source reports a 30 ms frame interval during RESET (PDF p. 109). It does not provide an exact transition time from that interval alone. Several visually different growth and dissolution modes are reported for different stacks; the material, electrode, environment and protocol matter.

Compare TEM cases rather than a universal filament

Reported case	Observation under its experiment	Interpretation and remaining limit
Ag cluster in evaporated SiO_2, electric field applied	A cluster disappears while another emerges nearby.	Dissolution and re-nucleation can mimic translation; frames do not show all intermediate processes.
Cu/WO_x/TiN, 10 cycles with SET compliance increased from 20 to 300 µA	HRS degraded as the protocol and contrast evolved.	The authors associate the changes with Cu deposition; compliance and cycle history change together, so this sequence alone is not a controlled single-variable proof.
Pt-Ir/NiO/Pt-Ir, forming with a 1 kΩ series resistor	An oxygen-poor region formed; subsequent RESET was not achieved.	The report attributes failure to oxygen lost to vacuum; thinning, current overshoot and environment restrict generalization.

These are distinct devices and measurement protocols (source PDF pp. 104, 110–113); do not transfer a frame interval or current limit between rows.

A counted EDS peak: a deliberately simple example

Suppose two equally exposed spectra use the same collection geometry and calibration. The measured Ag-peak window has 80 counts after SET and 50 after RESET; the estimated background is 20 counts in each window. Net counts are $80 - 20 = 60$ and $50 - 20 = 30$: the first is twice the second.

These are invented counts for arithmetic practice. A net-count ratio is **not** automatically a concentration ratio: thickness, detector efficiency, overlap, statistical uncertainty and local geometry matter. Compare reference regions and trace the image/spectrum acquisition to the same bias history before attributing a state change to Ag motion.

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CAFM: one sample, multiple forming locations

In one reported HfO_2/Zr/TiN sample, a PtIr-coated tip was used as the local top electrode at 12 selected positions. Two locations had a breakdown/forming voltage around 5 V and ten had values above 10 V; the reported subsequent switchable response appeared only at the two lower-voltage locations (source PDF p. 132).

For this sample and selection, $2/12 \approx 16.7\%$. It is not a device-yield specification or an estimate for every film: only 12 locations were tested, and tip geometry, local force, sampling, history and compliance affect the observation. Follow each location with a defined read/write/read sequence and an appropriate control.

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Scalpel SPM: identify which area is being reported

Example	Electrical preparation and display	What the maps support	Limit
A reported CBRAM cell	After device programming, the reported CBRAM sequence used ±2 V and 50 µA compliance; a 5 nA isosurface was displayed.	The reported displayed area changed from about 493 to 200 nm^2 across the oxide.	The authors estimated about 10% area overestimation for this particular analysis; threshold and tip contact affect the result.
A reported oxide-vacancy switching cell	A separate example used ±1 V and a 50 pA display threshold.	Sequential conductive maps help infer a depth-dependent feature.	Exposed oxide may re-oxidize; a different display threshold is not evidence of the same physical area.

Both reconstructions use sequential removal rather than simultaneous 3D imaging (source PDF pp. 140–142). The thresholds above are **display choices**, not forming currents or universal geometric boundaries.

References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Material cases: evidence and mechanisms

Four case families connect switching physics to measurements: SiO_2 conductive-bridge RAM, TaO_x RESET statistics, HfO_2/Ti aging, and two- and three-terminal atomic switches. Values below are either explicitly identified as historical observations or stated as hypothetical teaching models.

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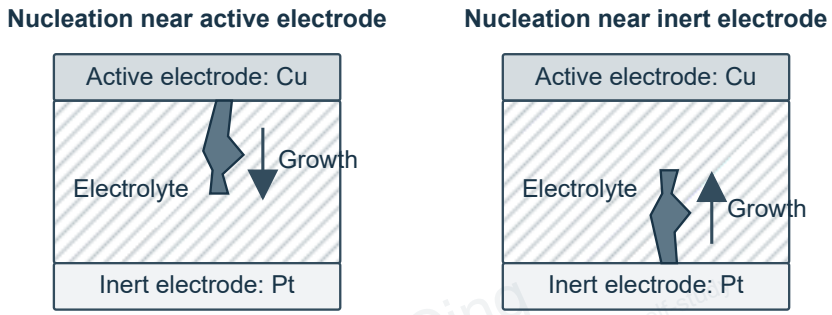
Compare mechanisms through their evidence

Case	Primary observation	A proposed explanation	What needs checking
SiO_2 CBRAM	switching and transport depend on bias and environment	metal ions participate in a conductive bridge	composition, polarity, transport and interfaces
TaO_x RESET	distributions differ with measurement and initial state	thermally assisted local rupture	current, temperature model and sample variability
HfO_2/Ti	the HRS/LRS window evolves during storage	oxygen redistribution and Ti oxidation	separate devices, aging protocol and depth-profile artifacts
Atomic switch	a conducting gap can be opened or closed	nucleation, growth and dissolution	field, pulse timing and alternative kinetics

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A conductive bridge is a process, not a single reaction

Conductive bridge: nucleation can differ
Schematic; reaction, supply and transport govern growth



Both scenarios are possible; polarity alone does not set direction.
Bridge formation also needs ion supply, reduction and connection.

For an illustrative Cu-active electrode, oxidation can supply ions: $\mathrm{Cu} \rightarrow \mathrm{Cu}^{+} + e^{-}$. A reduction step can return ions to metal: $\mathrm{Cu}^{+} + e^{-} \rightarrow \mathrm{Cu}$. Their locations and rates depend on electrode materials, interfacial kinetics, ion transport and electronic access.

Polarity alone does not determine morphology

Ion drift, diffusion, interfacial reduction and existing clusters can compete. Published material systems show growth toward either electrode or distributed nucleation; a voltage sign by itself cannot establish a universal growth direction. The return of HRS also need not remove every metallic atom.

To test a mechanism, align electrical switching with composition-sensitive imaging, repeat cycles, control humidity and temperature, and check whether the probe changes the state.

A diffusion scale is not a SET-time prediction

For ideal one-dimensional unbiased diffusion from a point, the mean-square displacement is $\langle x^2 \rangle = 2Dt$. For a deliberately hypothetical $D = 10^{-21}\,\mathrm{cm^2/s}$ and rms displacement $20\,\mathrm{nm} = 2 \times 10^{-6}\,\mathrm{cm}$,

$$t = \frac{(2 \times 10^{-6})^2}{2 \times 10^{-21}} = 2 \times 10^{9}\,\mathrm{s} \approx 63\,\mathrm{yr}.$$

This extrapolation excludes electric-field drift, interfacial reactions and thin-film pathways. It is not a device switching time or evidence identifying one mechanism.

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Field and humidity change the boundary conditions

Under a field, ionic flux has drift and diffusion contributions; a high-field empirical fit applies only within its measured range. Boundary reaction rates and available ions also matter.

In a particular reported Cu/SiO_2/Pt experiment, changing water-vapor conditions affected the redox and ion-population interpretation; this does not establish that humidity always increases ionic mobility. Compare matched samples and histories before attributing a change to one parameter.

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Field drift: a limited numerical comparison

For a hypothetical dilute, isothermal, singly charged carrier in the linear-response regime, use the Einstein relation $\mu = D/(k_B T/e)$ and drift speed $v_d = \mu E$. Take $D = 10^{-21}\,\mathrm{cm^2/s}$, $k_B T/e = 0.02585\,\mathrm{V}$, and $E = 10^5\,\mathrm{V/cm}$:

$$\mu = 3.868 \times 10^{-20}\,\mathrm{cm^2/(V\,s)}, \quad v_d = 3.868 \times 10^{-15}\,\mathrm{cm/s}.$$

The drift distance divided by v_d for $L = 20\,\mathrm{nm}$ is $5.17 \times 10^8\,\mathrm{s}$. This is an illustrative drift scale, not a first-passage prediction or a SET time. Linear response and the Einstein relation are explicit assumptions; the field value by itself does not establish them. For a hopping picture, $qaE \ll k_B T$ is one condition to check. Constant field, dilute independent carriers and these relations may fail in a real switching gap.

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What changing humidity can and cannot identify

In a uniform drift-only illustrative model $J = qn\mu E$. The same change in J could result from a change in mobile-carrier concentration n, mobility μ, or field E; interface reactions and diffusive flux add further ambiguity. A current change alone cannot separately identify n and μ.

At fixed geometry and temperature, measure a concentration-sensitive signal and independently test transport across several fields and matched humidity histories; the result remains conditional on the chosen transport and contact model.

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A simple impedance model and its limits

Suppose a resistance R and capacitance C are in parallel. Their admittance is $Y = 1/R + j\omega C$; hence

$$Z = \frac{R}{1 + j\omega RC}, \qquad -\operatorname{Im} Z = \frac{\omega R^2 C}{1 + (\omega RC)^2}.$$

The imaginary component reaches its maximum at $\omega RC = 1$. Fitting this equivalent circuit describes a response under its assumptions; it does not uniquely reveal a filament's shape or composition.

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Work through the RC scale

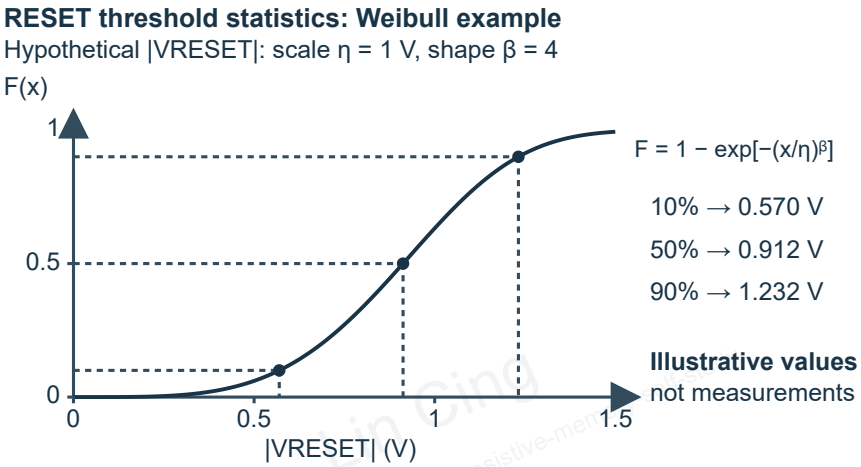
Let $R = 1\,\mathrm{M\Omega}$ and $C = 10\,\mathrm{pF}$ in the parallel model. Then $\tau = RC = 10\,\mu\mathrm{s}$ and

$$f_{\mathrm{peak}} = \frac{1}{2\pi RC} \approx 15.9\,\mathrm{kHz}.$$

Measure over a sufficient frequency range and check lead and electrode contributions. More than one physical arrangement may fit the same spectrum.

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RESET is a distribution, not one universal number



η: 63.2% quantile. Report binning, correlation and censoring.

One historical Ta/TaOx/Pt study used an approximately 18 nm oxide and repeated quasi-DC cycles on a device. SET compliance, RESET stop voltage, voltage polarity, read voltage and cycle history are part of that particular dataset, not universal settings.

For a negative-polarity RESET experiment, define the nonnegative variable $X = |V_{\mathrm{RESET}}|$ before fitting. A peak-current criterion or another chosen criterion must be stated explicitly.

One TaOx RESET study: protocol before inference

In the specific historical Ta/TaOx/Pt experiment, the oxide was about 18 nm; approximately 1,000 quasi-DC cycles were recorded on the same device. SET used positive bias with $4\,\mathrm{mA}$ compliance, RESET swept negative to a $-1.5\,\mathrm{V}$ stop, and R_{ON} was read at $0.1\,\mathrm{V}$; the voltage-step time was $1\,\mathrm{ms}$. RESET voltage was defined through the current peak.

These settings and the chosen event definition describe that dataset. Cycle history and resistance binning affect the observed distribution and are not independent population statistics.

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Six resistance bins under a simple assumption

Assume six hypothetical resistance bins and a common peak magnitude $|V_{\mathrm{RESET}}| = 0.9\,\mathrm{V}$. If the instantaneous peak is approximated by $I_{\mathrm{peak}} = |V|/R_{\mathrm{ON}}$:

R_{ON} (Ω)	100	150	200	250	300	400
I_{peak} (mA)	9	6	4.5	3.6	3	2.25

For example, $0.9/150 = 0.006\,\mathrm{A} = 6\,\mathrm{mA}$. These hypothetical fixed-voltage, Ohmic values are not a reconstruction of the published Weibull distribution. In this model the current is inversely proportional to resistance; evolving resistance or a nonconstant peak voltage invalidates the simple calculation.

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Weibull model: CDF and quantile

For $x \geq 0$, shape $\beta > 0$, and scale $\eta > 0$, a two-parameter Weibull model is

$$F(x) = 1 - e^{-(x/\eta)^{\beta}}, \qquad x_p = \eta[-\ln(1-p)]^{1/\beta}.$$

At $x = \eta$, $F(\eta) = 1 - e^{-1} \approx 0.632$. This is a model definition; fitting data does not establish that all devices obey it. Use magnitude consistently and treat censored observations explicitly.

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Weibull worked example

Assume $\eta = 1\,\mathrm{V}$, $\beta = 4$. At $0.8\,\mathrm{V}$, $(0.8)^4 = 0.4096$ and $F = 1 - e^{-0.4096} \approx 0.3361$.

The 10%, 50%, and 90% quantiles are approximately 0.5697, 0.9124, and $1.2318\,\mathrm{V}$. They describe the assumed distribution, not a measured performance guarantee.

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How to fit without manufacturing certainty

For $x > 0$ and $0 < F < 1$, $\ln[-\ln(1 - F)] = \beta \ln x - \beta \ln \eta$. A Weibull plot can reveal departures from a straight line; it cannot validate a mechanism on its own.

Report the sample count, device count, cycle dependence, selection or resistance binning, censoring and confidence intervals. Repeated cycles from one device are not automatically independent devices.

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A thermal RESET model is conditional

Assume steady local heating with constant R and thermal resistance R_{th} to a bath at T_0:

$$P = \frac{V^2}{R}, \qquad T = T_0 + R_{\mathrm{th}} P.$$

This lumped model excludes thermal transients, spatial temperature gradients, evolving resistance and nonthermal mechanisms. Agreement with it can support a hypothesis but cannot uniquely establish it.

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Thermal estimate and scaling

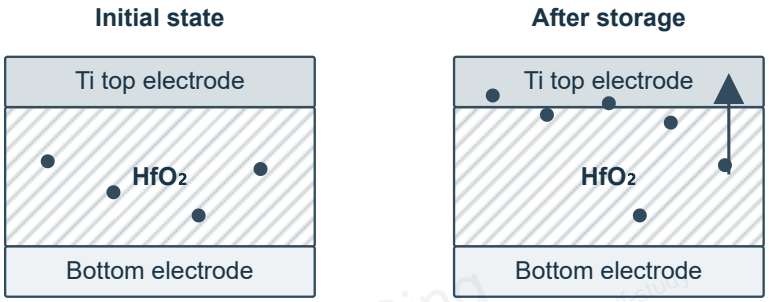
Take hypothetical $R = 150\,\Omega$, $V = 0.9\,\mathrm{V}$, $R_{\mathrm{th}} = 3 \times 10^4\,\mathrm{K/W}$. Then $P = 0.81/150 = 5.4\,\mathrm{mW}$ and $\Delta T = 162\,\mathrm{K}$.

If RESET requires the same critical temperature rise, $V_{\mathrm{RESET}}^2 = \Delta T_c R / R_{\mathrm{th}}$. A nearly constant RESET voltage across different R follows only if ΔT_c and R/R_{th} remain approximately fixed.

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Aging: a measurement protocol is part of the result

HfO_2/Ti ageing: material distribution and evidence
Conceptual illustration; not a measured concentration profile



- Oxygen-related species; arrow shows one possible change.

XPS is indirect; sputtering and sample variation affect interpretation.
Report storage time, structure and measurement conditions.

The cited HfO_2/Ti work studied Pt/HfO_2/Ti/Pt stacks and compared nominal Ti thicknesses, aging intervals and electrical states. The published curves concern specified batches and read conditions; they are not a lifetime guarantee for all HfO_2 cells.

Interpret a changing memory window

For a specified group, a reported HRS/LRS ratio fell from roughly 10^4 to 10^2 over about two weeks: the ratio is 100 times smaller. That does not determine the absolute HRS or LRS values individually.

Depth-profile XPS can support changes in elemental or chemical distributions, but sputtering and sample preparation may alter the profile. It is not a real-time movie of oxygen motion in the same operating cell.

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An aging comparison must preserve the sample identity

The reported stack was Pt/HfO_2 (about 10 nm)/Ti/Pt. Compare nominal Ti thicknesses 10, 30 and 100 nm; the approximately two-week fall from 10^4 to 10^2 concerns the 30 nm group under its particular protocol. The study first applied 15 cycles and then averaged five cycles per cell; storage comparisons used separate fresh devices from the same batch.

Step	What is compared	What the comparison does not establish
Prepare	Same-batch devices; specified Ti thickness and storage time	One device followed continuously unless explicitly measured
Condition and read	Forming and cycles under specified voltage/compliance and read convention	Equal initial states or identical HRS and LRS separately
Profile	Day 1 versus day 60 XPS depth profiles	Undisturbed, real-time motion during switching

At about 60 days, reported ratios for nominal 10/30/100 nm groups were approximately $5 \times 10^4/10^2/10$, respectively, with device and protocol dependence. Read conventions and some labels differ across adjacent source descriptions; do not merge ambiguous conditions without checking the original experiment.

Keep observations and explanations separate

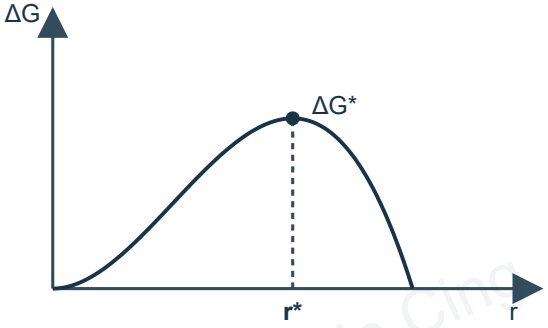
Compare matched fresh and aged devices, specify the Ti thickness, environment, preconditioning, read polarity and uncertainty. A TiO_2 insertion layer is a distinct device structure, not a continuation of a bare-Ti sample.

Reported figure labels and adjacent descriptions may differ in thickness or read conventions. Do not merge uncertain conditions: check the original experiment before reproducing or comparing its numeric values.

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A nucleation model for an atomic switch

Heterogeneous nucleation: surface versus volume energy
Model: fixed interfacial energy, driving force and cap shape



Small: surface cost
Large: volume gain

$r^* = 2\gamma / |\Delta g_v|$

$\Delta G^* = f(\theta)\, 16\pi\gamma^3 / [3|\Delta g_v|^2]$

Barrier ≠ rate: prefactors and transport also matter.
Nanoscale clusters or dynamic fields can invalidate the model.

In a classical spherical-cap approximation, let $\gamma > 0$ be surface energy, $|\Delta g_v| > 0$ the magnitude of bulk driving free energy per volume, and $0 < f(\theta) \leq 1$ a geometric factor. Then

$$\Delta G(r) = f(\theta)\left(4\pi r^2\gamma - \frac{4\pi}{3}r^3|\Delta g_v|\right).$$

Derive the critical radius and barrier

For fixed $f(\theta) > 0$, differentiate with respect to r: $d\Delta G/dr = f(8\pi r\gamma - 4\pi r^2|\Delta g_v|)$. The nonzero stationary radius is

$$r^* = \frac{2\gamma}{|\Delta g_v|}, \qquad \Delta G^* = f(\theta)\frac{16\pi\gamma^3}{3|\Delta g_v|^2}.$$

It is a maximum under this model. Classical continuous geometry and constant material properties may be poor approximations for a cluster only a few atoms wide.

Nucleation example: calculate before interpreting

Assume $\gamma = 0.03\,\mathrm{J/m^2}$, $|\Delta g_v| = 3 \times 10^7\,\mathrm{J/m^3}$ and $f = 0.1$. Then $r^* = 2 \times 10^{-9}\,\mathrm{m} = 2\,\mathrm{nm}$ and $\Delta G^* \approx 5.03 \times 10^{-20}\,\mathrm{J} \approx 0.314\,\mathrm{eV}$.

At 300 K, use $k_B T \approx 0.02585\,\mathrm{eV}$ and $e^{-\Delta G^*/k_B T} \approx 5.3 \times 10^{-6}$. This factor alone is not a nucleation rate: the attempt frequency, available sites and nonequilibrium transport are not given.

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Time-resolved switching needs a defined event

A voltage edge applied to a device and its parasitic capacitance can produce $I_C = C\,dV/dt$ before a conducting bridge forms. A later jump in conduction current is a different event.

Define SET time relative to the applied pulse and a stated onset criterion; bandwidth, trigger delay and capacitance affect the measurement. A published nanosecond value belongs to its particular structure and measurement chain.

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Separate a capacitive peak from a conducting event

Assume $C = 0.2\,\mathrm{pF}$ and $dV/dt = 1\,\mathrm{V/ns}$. Before conduction, the ideal capacitive current is $0.2\,\mathrm{mA}$; seeing this at $0.5\,\mathrm{ns}$ does not establish SET.

If the stated onset of conduction occurs at $1.2\,\mathrm{ns}$ after pulse start, the defined SET time is $1.2\,\mathrm{ns}$. Device energy requires $\int V(t) I_{\mathrm{device}}(t)\,dt$ with a defined interval and treatment of parasitic energy; a peak alone is insufficient.

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A slope that does not identify a mechanism

Consider two deliberately simplified models on the voltage window $1\text{–}2\,\mathrm{V}$. A nucleation ansatz has $t = A\exp[\Delta G(V)/(k_B T)]$, with $\Delta G(V) = \Delta G_0 - 0.05\,\mathrm{eV}\,(V/1\,\mathrm{V})$, $k_B T = 0.025\,\mathrm{eV}$, and $\Delta G_0 > 0.10\,\mathrm{eV}$ so the barrier remains positive throughout this interval. Choose the combined prefactor so that $\ln(t/\mathrm{ns}) = 6 - 2(V/1\,\mathrm{V})$.

Alternatively, an assumed high-field migration law $v = v_0 \exp[eaV/(2k_B T L)]$, with $a = 0.3\,\mathrm{nm}$, $L = 3\,\mathrm{nm}$, and $L/v_0 = e^6\,\mathrm{ns}$, gives the same expression for $t = L/v$. Each law has its own assumptions; neither is established by the example.

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Evaluate competing timing models

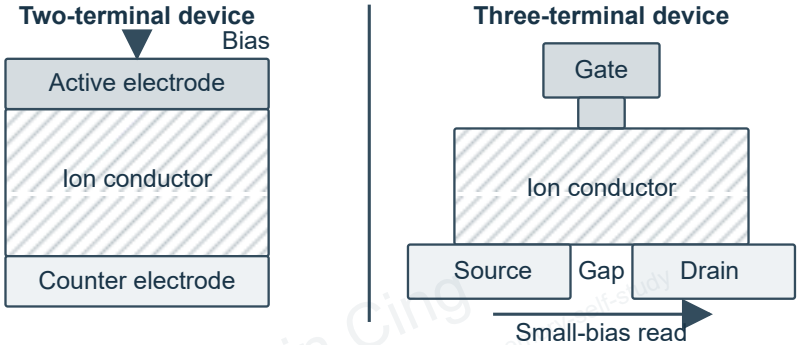
For either assumed law, $V = 1$, 1.5, and $2\,\mathrm{V}$ yield $t = e^4$, e^3, and $e^2\,\mathrm{ns}$: approximately 54.6, 20.1, and $7.39\,\mathrm{ns}$.

The identical fitted slope in this constructed example cannot distinguish mechanisms. Compare temperature, gap thickness, composition and time-resolved electrical/structural observations while considering model validity, parasitics and parameter uncertainty. Do not extrapolate a limited high-field fit to zero bias.

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Two terminals and three terminals answer different questions

Atomic switches: two-terminal and three-terminal
Schematics; materials and bias depend on the device



Connect source–drain while avoiding a gate short.
Ion motion and nucleation depend on material, bias and interfaces.

A two-terminal switch shares the applied voltage between the active electrode, electrolyte and counterelectrode. In a three-terminal example, a gate controls metal-ion supply while a smaller source–drain bias probes the gap. The conducting source–drain path must be distinguished from an unwanted short to gate.

Do not confuse a fit with a unique mechanism

Different kinetic models can fit a limited voltage dependence of switching time. Check their parameter values, geometry, temperature, pulse bandwidth and sensitivity to alternative assumptions. A fitted effective hopping distance larger than an atomic spacing may challenge one model, but a slope alone does not prove the remaining model.

As a final check, separate observed waveform, fitted model, inferred material change and untested predictions.

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Programming current and multilevel states

For a reported Cu/SiO_2 CBRAM family, ON resistance decreases as SET compliance current rises; a fitted form is $R_{\mathrm{ON}} = A / I_{\mathrm{prog}}^{n}$, with exponent near one **for that sample and fit range**. Compliance limits the SET current; it does not guarantee an identical filament geometry on every cycle.

Multiple levels are useful only when the read-current distributions remain distinguishable after programming variation, noise and retention drift. Four labels encode at most two bits per cell; overlapping distributions or time-dependent drift can reduce usable information.

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Trace three hypothetical current levels

Assume the special case $n = 1$ and $A = 0.5\,\mathrm{V}$; these are teaching values, not reported measurements. For $I_{\mathrm{prog}} = 5, 10, 20\,\mu\mathrm{A}$, the model yields $R_{\mathrm{ON}} = 100, 50, 25\,\mathrm{k\Omega}$.

At a non-disturbing $0.1\,\mathrm{V}$ Ohmic read, the corresponding currents are $1, 2, 4\,\mu\mathrm{A}$. These are three nominal levels; to choose thresholds, measure each level's full distribution under the intended read and waiting conditions.

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A conductance scale is not a geometry test

For spin-degenerate ideal channels, $G_0 = 2e^2/h \approx 77.5\,\mu\mathrm{S}$, and $1/G_0 \approx 12.9\,\mathrm{k\Omega}$. A resistance near this scale is suggestive of atomic-scale transport, but $G = G_0 \sum_i T_i$ also allows different channel transmissions to give the same conductance.

The book discusses filament widening, multiple paths and a residual gap as interpretations in different regimes. A resistance threshold by itself cannot prove whether a filament is complete, measure its width, or identify a unique atomic configuration.

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Multilevel retention is a distribution over time

For one reported PECVD Cu/SiO_2 device study, both HRS and LRS were reported to remain near their programmed values for over $10^4\,\mathrm{s}$ at $150\,^\circ\mathrm{C}$ under that experiment's conditions. This does not establish ten-year retention at another temperature or a bound on an unmeasured tail.

A concentration gradient and interfacial reactions can destabilize a metallic path; high temperature or residue can also change HRS in a particular system. Compare read distributions at stated times, temperatures, read bias and sampling intervals, including failed or censored cells.

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Irradiation: match cell state, dose and sampling

The cited Cu/SiO_2 CBRAM study used stepwise $^{60}\mathrm{Co}$ gamma exposure up to $7.1\,\mathrm{Mrad(SiO_2)}$. Virgin measurements included three $10 \times 10\,\mu\mathrm{m}^2$ and three $100 \times 100\,\mu\mathrm{m}^2$ cells, with sweeps limited to $100\,\mathrm{mV}$ to reduce electrode oxidation during this measurement.

For switched-state distributions, ten $10 \times 10\,\mu\mathrm{m}^2$ cells were cycled ten times at different doses. Similar plotted HRS/LRS distributions support limited sensitivity within that tested set, not universal radiation immunity or an upper bound on rare failures.

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Irradiation and multilevel programming

The same chapter reports three $10 \times 10\,\mu\mathrm{m}^2$ cells programmed over a range of compliance currents before and after the $7.1\,\mathrm{Mrad(SiO_2)}$ exposure; the plotted resistance–current curves were similar under those conditions.

The dose unit includes the reference material: do not silently replace $\mathrm{Mrad(SiO_2)}$ with a different material's absorbed-dose value. Compare the whole state distribution, dose history, read bias and repeated-cell dependence; three cells do not quantify low-probability failures.

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Humidity-controlled cyclic voltammetry: a separate experiment

One reported Cu/Ta_2O_5/Pt cyclic-voltammetry study varied RH from 5% to 85% using dry and humidified nitrogen and used a $100\,\mathrm{mV/s}$ sweep for the plotted RH comparison. With increasing RH, oxidation and reduction peaks became clearer; the reported reduction-peak position changed up to around 22% RH and showed little further change at higher RH in that comparison.

The inferred ion concentration increased with RH; the fitted diffusion coefficient initially decreased and then increased above roughly 50% RH. These are model-dependent estimates for this CV study, not measurements assigned automatically to the separate film-density switching study below. Several transport and interfacial mechanisms can coexist.

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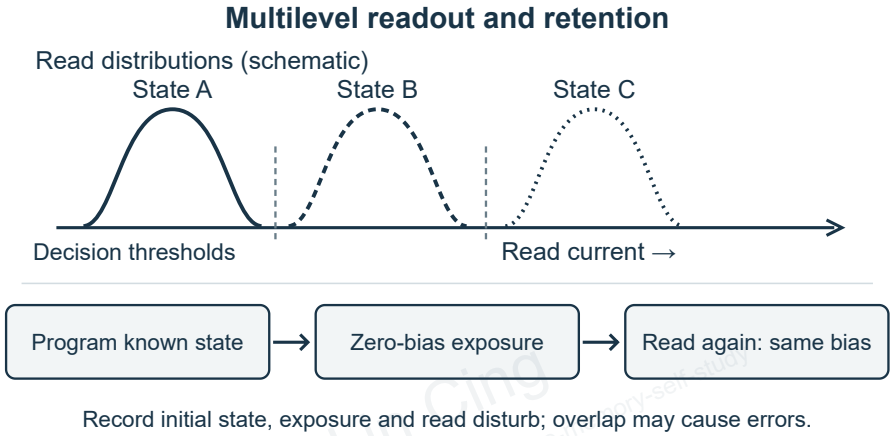
Film density and switching: a separate comparison

In a distinct reported Cu/Ta_2O_5/Pt comparison, films had densities of 7.6 and $7.1\,\mathrm{g/cm^3}$, and forming preceded the I–V measurement at each RH. The example I–V curves include 22% and 85% RH; the $7.6\,\mathrm{g/cm^3}$ sample showed nonvolatile switching under tested RH variation. The $7.1\,\mathrm{g/cm^3}$ sample changed from nonvolatile below 50% RH to volatile above 50%, with recovery when RH fell below 50%.

The authors proposed proton-related conduction as one explanation for the latter behavior and stated that further investigation was needed. Density, deposition and moisture history are coupled; these results do not establish a universal 50% threshold or identify the carrier by I–V shape alone.

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Multilevel readout and retention



Read each programmed state at the same bias before and after the specified zero-bias exposure. Distribution overlap can cause errors even when the nominal current levels differ.

Interpret each evidence chain

In situ plasmon-enhanced spectroscopy of Ag/SiO_2/Au can track spectral changes during switching and support residual particles and changing growth locations. A spectrum is not a direct image: electrode optical response, field enhancement and alternative structures affect interpretation.

For virgin Cu/SiO_2 devices, a parallel-RC fit and area-dependent R and C were reported; a programmed $500 \times 500\,\mu\mathrm{m}^2$ cell spanning about $230\,\Omega$–$55\,\mathrm{k}\Omega$ retained approximately the virgin fitted capacitance. Localized conduction is a plausible interpretation, but equivalent-circuit fits alone cannot identify a unique microscopic path.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Perovskite switching: interfaces and correlated oxides

Perovskite oxides can change resistance through interface barriers, defect redistribution or changes throughout a conducting layer. This unit separates observations from mechanism hypotheses and treats each equation as a model with explicit conditions.

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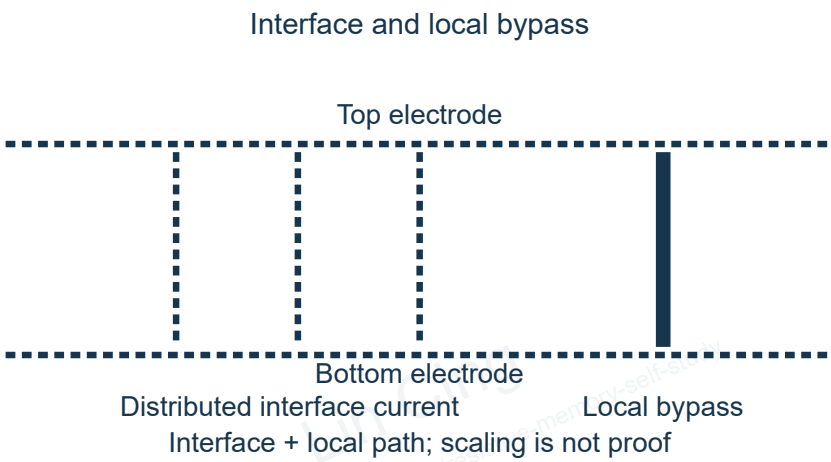
Start with material and measurement conditions

An ideal perovskite has an ABO_3 lattice. The geometric tolerance factor $t = (r_A + r_O)/(\sqrt{2}(r_B + r_O))$ uses ionic radii under a stated coordination convention. It is a structural guide, not a guarantee of phase stability or switching.

Stoichiometry, strain, grain boundaries, electrode chemistry, temperature and growth atmosphere can all change transport. Report the stack, polarity at a named electrode, sweep or pulse protocol, read bias and history.

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Interface and volume are different hypotheses



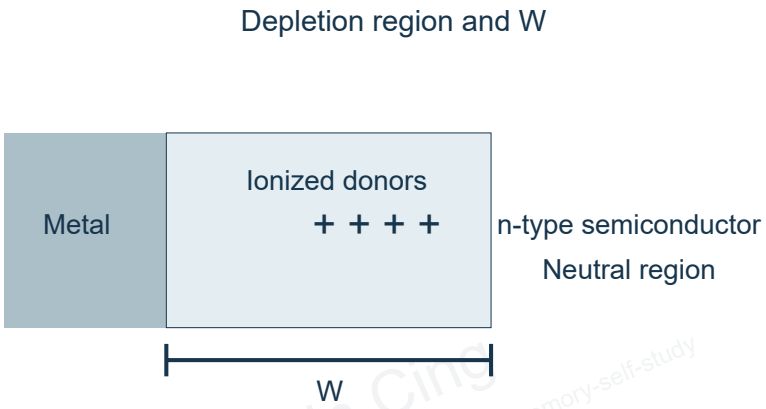
An interface hypothesis attributes a change mainly to a contact barrier or interfacial region. A volume hypothesis places an important change within a film or conducting path. Both can coexist, so terminal I–V curves alone rarely locate the change.

Competing paths complicate area scaling

Suppose a uniform interface has area-specific resistance $RA = 100\,\Omega\,\mu\mathrm{m}^2$. At fixed bias, temperature and state, its resistance is $R_i = RA/A$: for $A = 10$ and $40\,\mu\mathrm{m}^2$, $R_i = 10$ and $2.5\,\Omega$.

Now assume a separate area-independent $100\,\Omega$ path in parallel. The total resistance is $(1/R_i + 1/100)^{-1}$: 9.09 and $2.439\,\Omega$. A nearly inverse-area trend alone does not establish a unique mechanism.

A depletion approximation has prerequisites



Schematic: uniform donors, one-sided depletion

For a uniformly doped n-type semiconductor, complete donor ionization, negligible interface fixed charge, a one-sided depletion region and electrostatic potential drop $\phi > 0$, Poisson's equation gives $W = \sqrt{2\epsilon\phi/(qN_D)}$. The film must be thick enough for the assumed region. Real interfacial states and tunneling can invalidate this simplified picture.

Derive the width before using it

With constant charge density $\rho = qN_D$ over $0 < x < W$ and $E(W) = 0$, integrate $dE/dx = \rho/\epsilon$. The magnitude of the field is $|E(x)| = qN_D(W - x)/\epsilon$. The potential drop is the area under $|E|$:

$$\phi = \int_0^W |E(x)|dx = \frac{qN_D W^2}{2\epsilon}.$$

This is an equilibrium electrostatic approximation, not a switching-time model.

Work the depletion example

Take $\epsilon = 100\epsilon_0$, $\epsilon_0 = 8.854 \times 10^{-12}\,\mathrm{F/m}$, $\phi = 0.5\,\mathrm{V}$, $q = 1.602 \times 10^{-19}\,\mathrm{C}$ and $N_D = 10^{18}\,\mathrm{cm^{-3}} = 10^{24}\,\mathrm{m^{-3}}$.

The model gives $W \approx 74.3\,\mathrm{nm}$. If only N_D is quadrupled while the assumed potential and all other conditions are held fixed, W halves to $37.2\,\mathrm{nm}$. In an actual contact, ϕ and defect populations can also change.

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What can change a barrier?

Charged oxygen vacancies, trapped electrons, interface states, chemical reactions and polarization can each alter the electrostatic or chemical boundary. A thinner depletion region may increase tunneling; neither a particular polarity nor a monotonic improvement with doping follows universally from the material name.

One reported Nb-doped $SrTiO_3$ series changed dopant levels and showed disappearance of hysteresis at its highest specified doping. It illustrates a possible tradeoff, not a universal critical dopant level.

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Area, bias and temperature are evidence, not verdicts

Compare devices with several areas while controlling electrodes, temperature, voltage, state and cycle history. Test whether current density, resistance-area product or apparent activation changes systematically. A parallel path, varying contacts or current crowding can mimic simple scaling.

Use spatial mapping, spectroscopy, timing and controlled materials changes to distinguish competing hypotheses. Correlation alone does not identify a unique microscopic path.

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Space-charge-limited current: a restricted reference

In a planar trap-free, single-carrier model with ohmic injection, negligible equilibrium carriers and diffusion, constant mobility μ and permittivity ϵ, the current density is

$$J = \frac{9}{8}\epsilon\mu\frac{V^2}{d^3}.$$

The expression excludes contact-limited injection, substantial trapping, self-heating and field-dependent mobility. A measured log–log slope near two is insufficient by itself to prove this model.

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Derive the ideal quadratic law

Assume unipolar drift $J = qn\mu E$ and Poisson $dE/dx = qn/\epsilon$. Eliminating n yields $E\,dE/dx = J/(\epsilon\mu)$. With the ideal injecting boundary $E(0) = 0$, integrate to obtain $E(x) = \sqrt{2Jx/(\epsilon\mu)}$.

Integrating $V = \int_0^d E(x)dx$ and solving for J gives the $9/8$ law. The boundary and constant-parameter assumptions are part of the derivation.

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A slope can move without a new phase

Let an illustrative positive-bias current be $I = aV + bV^2$ with $a = 1\,\mu\mathrm{A/V}$ and $b = 1\,\mu\mathrm{A/V^2}$. Its local log–log slope is

$$\gamma = \frac{d\ln I}{d\ln V} = \frac{a + 2bV}{a + bV}.$$

At $V = 0.1, 1, 10\,\mathrm{V}$, $\gamma \approx 1.091, 1.5, 1.909$. The smooth crossover approaches two even though the chosen model contains two contributions throughout.

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Four transport models: distinct hypotheses

Model	Idealized regime	Test often used, with caveats
Schottky emission	thermionic injection over an interface barrier with image-force lowering	at fixed T, compare $\ln(J/T^2)$ with $\sqrt{E}$; independently check temperature dependence
Fowler–Nordheim (FN)	tunneling through an approximately triangular barrier under sufficiently high field	compare $\ln(J/E^2)$ with $1/E$ and test barrier/field assumptions
Poole–Frenkel (PF)	thermally assisted emission from bulk traps, with field-lowered Coulomb barrier	at fixed T, compare $\ln(J/E)$ with $\sqrt{E}$ and test temperature and thickness
Trap-free SCLC	ohmic injection and bulk unipolar space charge, with constant mobility and permittivity	check $J \propto V^2/d^3$ under matched conditions, not slope alone

These are simplified diagnostics, not mutually exclusive labels automatically assigned by a straight fitted line.

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Model comparisons need actual fields

For ideal Schottky image-force lowering, $\Delta\phi_S = \sqrt{qE/(4\pi\epsilon)}$ expressed in volts; divide the barrier energy by q when using electronvolts. An illustrative current takes $J \propto T^2 \exp[-(\phi_B - \Delta\phi_S)/(k_B T/q)]$. The corresponding ideal PF Coulomb-trap barrier lowering is $\Delta\phi_{PF} = \sqrt{qE/(\pi\epsilon)}$, twice as large for the same assumed E and ϵ.

The relevant local field need not equal external V/d: unknown interface drops, geometry, heating and series resistance can spoil either transform. Different mechanisms can produce an approximately straight graph over a narrow range.

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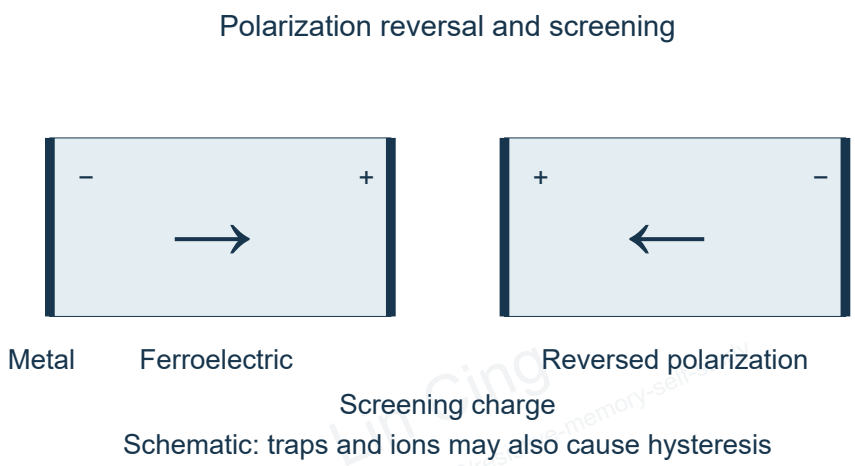
Work a conditional Schottky comparison

Assume a fixed barrier, temperature and prefactor, with $T = 300\,\mathrm{K}$, $k_B T/q = 0.02585\,\mathrm{V}$, $\epsilon_r = 9$ and local fields $E_1 = 0$, $E_2 = 10^7\,\mathrm{V/m}$. With $\epsilon = \epsilon_r \epsilon_0$, the ideal Schottky barrier lowering changes from zero to $\Delta\phi_S \approx 0.0400\,\mathrm{V}$.

Holding all other factors fixed, the model's exponential factor changes by $\exp(0.0400/0.02585) \approx 4.70$. The ideal PF lowering at E_2 is $\Delta\phi_{PF} \approx 0.0800\,\mathrm{V}$, so its corresponding exponential factor changes by about 22.1. **These are barrier factors**, not the complete PF current ratio: its field-dependent prefactor must also be accounted for, and the complete PF current at $E_1 = 0$ is zero in the usual ideal model. Neither comparison identifies an actual mechanism.

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Traps and polarization can imitate one another



Trapped charge and ferroelectric polarization can both modify an interface field. Ferroelectric switching changes polarization; a low-disturbance resistance read does not automatically reverse it. Sweep hysteresis alone cannot rule out ionic migration or charge trapping.

Design a discriminating experiment

Compare write polarity, pulse amplitude and duration, read-disturb response, waiting time and temperature. Independently measure polarization when possible, and test electrodes or layers that alter ion exchange. Match the measurement history and assess whether the probes themselves modify the state.

Different time scales provide useful evidence, but an observed correlation with polarization is not sufficient to exclude all other mechanisms.

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Defect accounting precedes a scaling law

In Kröger–Vink notation, a doubly positive oxygen vacancy is $V_O^{\bullet\bullet}$ relative to the lattice and an electron is e'. For the illustrative equilibrium reaction

$$O_O^{\times} \rightleftharpoons \tfrac{1}{2}O_2(g) + V_O^{\bullet\bullet} + 2e',$$

charge neutrality $n = 2[V_O^{\bullet\bullet}]$ holds only if these are the relevant compensating species. Charged acceptors, holes or fixed charges require an expanded balance.

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Derive an oxygen-pressure exponent

Assume an ideal dilute equilibrium, constant lattice-site activity, constant temperature, and the reaction and charge balance above. The mass-action condition is $[V_O]n^2p_{O_2}^{1/2} = K$, so $n = 2[V_O]$ gives

$$n^3 \propto p_{O_2}^{-1/2}, \qquad n \propto p_{O_2}^{-1/6}.$$

Raising p_{O_2} by 64 therefore halves n. Only if electron mobility is also unchanged does the electron conductivity halve. Non-equilibrium write pulses and other defects invalidate the simple exponent.

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Humidity changes a boundary, not a universal setting

One standard idealized hydration reaction is

$$H_2O + V_O^{\bullet\bullet} + O_O^{\times} \rightleftharpoons 2OH_O^{\bullet}.$$

A particular Pt/$SrTiO_3$/Pt study reported stable switching at an intermediate humidity but not at its tested dry and saturated conditions. This sample-specific observation does not mean more moisture is always beneficial, nor does it identify a unique proton concentration.

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A humidity experiment needs matched controls

Specify sample preparation, relative humidity, temperature, bias, sweep history and waiting time. Compare dry, intermediate and humid environments on matched devices, including recovery after returning to the initial environment. Separate immediate surface effects from irreversible aging or electrode chemistry.

A changed I–V loop is an observation. Identifying ionic species or their concentration requires additional chemical and spatial evidence.

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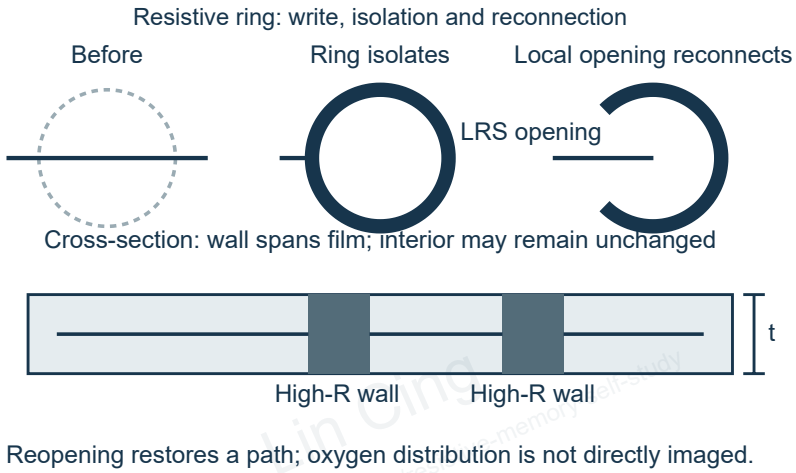
Strongly correlated oxides: a volume hypothesis

The electrical state of a correlated oxide can depend on electronic carrier concentration, lattice structure and temperature. Oxygen exchange may change local composition and shift a metal–insulator transition (MIT); this is a material- and preparation-dependent hypothesis, not an automatic explanation for every perovskite.

An observed transition temperature is extracted with a stated temperature sweep and resistance criterion. If one sample changes from 315 K to 200 K under a specified treatment, the shift is $-115\,\mathrm{K}$; it is not a universal device specification.

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A written ring can isolate an unwritten interior



Reopening restores a path; oxygen distribution is not directly imaged.

Suppose a continuous high-resistance wall crosses the film thickness and encloses an unmodified interior. A lateral current path from outside to inside is interrupted even when the interior itself has not been written. A local low-resistance opening through the wall can reconnect the regions.

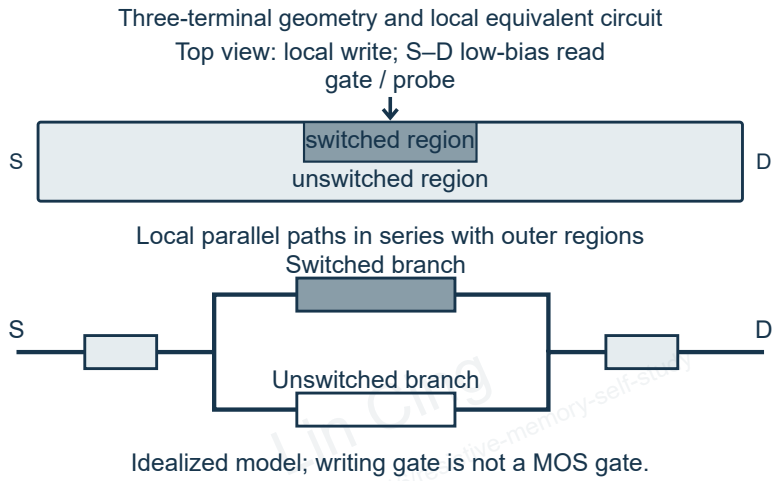
In a reported LSMO experiment, a $-6\,\mathrm{V}$ ring left a path between interior and exterior, whereas a $-10\,\mathrm{V}$ written ring electrically isolated the unwritten interior; a local $+10\,\mathrm{V}$ rewrite restored a path. These are the signs and conditions of that particular experiment. The result supports a continuous-wall hypothesis but does not directly measure oxygen concentration or prove a uniform wall.

Ring experiment: controls and inference

Keep the tip/sample voltage convention, film thickness, scan path, humidity and temperature fixed; map the interior before and after writing the ring without deliberately writing the interior. Compare the -6 and $-10\,\mathrm{V}$ rings and the local $+10\,\mathrm{V}$ reconnecting stroke. Repeat on control areas and monitor probe condition.

The low-humidity protocol limits an environmental variable; it does not remove every chemical or thermal effect. A complementary depth- or composition-sensitive measurement is needed before interpreting the barrier as a particular oxygen profile.

Three-terminal geometry changes the question



A source–drain bridge is read laterally while a gate or probe changes a selected region. This separation is useful for testing transport, but does not make the device an ordinary MOSFET. Gate current, local chemistry, contacts and alternate paths must be measured or controlled.

A bridge with a partial high-resistance band

Assume a uniform rectangular initial film with $L = 200\,\mu\mathrm{m}$, $W = 20\,\mu\mathrm{m}$ and total $R_0 = 1\,\mathrm{M\Omega}$. A band of length $\ell = 10\,\mu\mathrm{m}$ has half its width changed to resistivity 100 times the initial value. Treat the two half-width parts of the band as parallel, and the remaining length as series; ignore contact resistance and lateral current spreading.

The unchanged outside is $0.95R_0$. The band's two branch resistances are $0.1R_0$ and $10R_0$, giving $R_{\mathrm{band}} = 0.09901R_0$ and $R \approx 1.04901\,\mathrm{M\Omega}$.

Full-width change has a different signature

Under the same simplified model, a full-width band of length $20\,\mu\mathrm{m}$ with resistivity ratio 1000 gives

$$R/R_0 = 0.9 + 1000(0.1) = 100.9.$$

Thus $R \approx 100.9\,\mathrm{M\Omega}$. A total resistance alone still cannot uniquely recover the switched geometry: the conclusion depends on assumed uniformity, contacts and neglected current spreading.

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Read margin is about distributions

At an ideal fixed $1\,\mathrm{V}$ read bias, $1\,\mathrm{M\Omega}$ HRS and $10\,\mathrm{k\Omega}$ LRS give $1\,\mu\mathrm{A}$ and $100\,\mu\mathrm{A}$. These are single-state examples, not measured yield.

To design a threshold, compare state distributions, sense noise and device-to-device or cycle variation, and check read disturbance. A single large nominal ratio does not establish long-term reliability.

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A practical evidence checklist

1. State the electrode, voltage sign, pulse or sweep, read bias and temperature.
2. Specify the model and assumptions behind each fit, including any parallel paths.
3. Test spatial, chemical and temporal predictions against controls.
4. Report competing explanations and what the experiment does not resolve.

Interface, filament and volume changes can coexist; distinguish what is measured from what is inferred.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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$SrTiO_3$: thin films, single crystals and time-dependent switching

$SrTiO_3$ (STO) is a model oxide in which mobile defects, interfaces and electronic carriers can interact. A switching curve is an observation; locating its cause requires controlled evidence from geometry, time, temperature and contacts.

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A map of the two cases

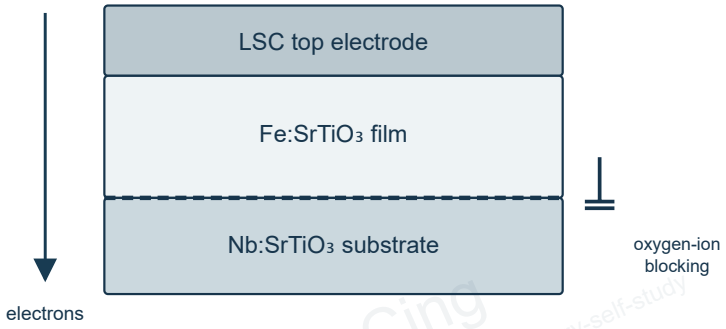
The thin-film case combines impedance spectroscopy, oxygen isotope exchange and voltage sweeps. The single-crystal cases contrast defect redistribution with an interface-barrier interpretation. Their materials, temperatures and voltage protocols differ and must not be merged into one device specification.

We will calculate characteristic times, ionic conductivity, effective barrier changes and pulse overlaps, then examine what each calculation does—and does not—identify.

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A thin-film stack with two transport channels

High-temperature film: electronic path and ionic boundary



Specific high-temperature samples (about 350–750 °C); boundaries depend on the system

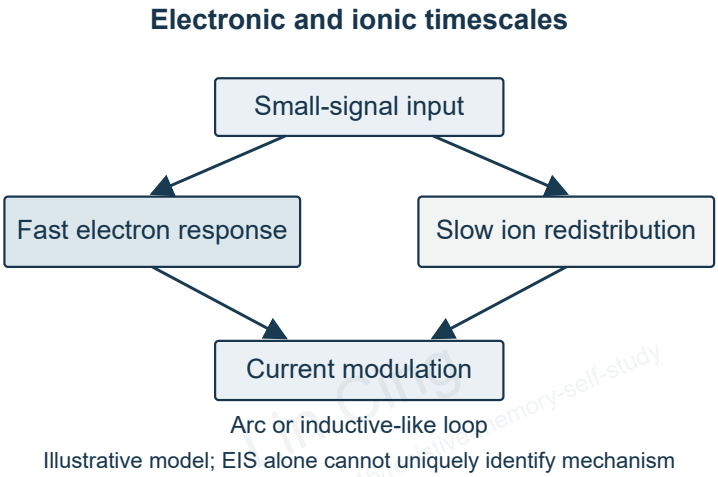
One reported high-temperature stack used 0.37 mol% Fe-doped STO film on a 0.5 wt% Nb-doped STO substrate with an LSC top electrode. Film thickness was 100–400 nm and top-electrode diameter 100–300 µm; the substrate is treated as electronically conductive but oxygen blocking under the specified model conditions. This boundary is an assumption to test for a new stack.

How geometry helps identify a response

For a uniform film of thickness d, area A and conductivity σ, $R = d/(\sigma A)$ and $C = \epsilon A/d$. Increasing d at fixed A, material and temperature should increase the bulk resistance and decrease the geometric capacitance.

Joint consistency with both trends strengthens a bulk assignment. It does not rule out an interface contribution: a contact resistance in series, nonuniform current or altered microstructure can change the result.

Small-signal impedance and characteristic time



Impedance $Z(\omega)$ is the ratio of small sinusoidal voltage to current around a specified operating point. For an ideal parallel RC element, $Z = R/(1 + i\omega RC)$; its loss peak occurs at $\omega = 1/(RC)$.

One circuit can fit multiple microscopic models. Identify temperature, DC bias, perturbation amplitude, frequency range and prior conditioning before assigning an arc to a mechanism.

Derive and use the ideal RC loss peak

The imaginary part is $-\operatorname{Im} Z = R(\omega RC)/(1 + (\omega RC)^2)$. Setting its derivative with respect to ωRC to zero gives $\omega RC = 1$ and $f_{peak} = 1/(2\pi RC)$.

For a hypothetical $R = 1\,\mathrm{M\Omega}$ and $C = 1\,\mu\mathrm{F}$, $\tau = RC = 1\,\mathrm{s}$ and $f_{peak} = 0.159\,\mathrm{Hz}$. An equivalent circuit is a model of response, not a direct image of oxygen vacancies.

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Why an inductive loop is not necessarily an inductor

The measured response can include a slow internal state x that changes electronic current. Linearize $I(V, x)$ and the state dynamics around the DC operating point: the delayed contribution can alter the phase and produce an apparent inductive loop.

An ionic redistribution model may explain such a feature, but the loop alone neither proves a physical inductance nor uniquely identifies a vacancy species. Check instrument artifacts, stability, bias dependence and alternative dynamic models.

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A minimal dynamic-state impedance model

Let $I = G_0 V + g x$ and $\tau \dot{x} = -(x - aV)$, linearized about an operating point with constant G_0, g, a, τ. For a sinusoidal perturbation with the $e^{i\omega t}$ convention,

$$Y(\omega) = \frac{\delta I}{\delta V} = G_0 + \frac{ga}{1 + i\omega\tau}.$$

This illustrative model is stable and passive under the sufficient conditions $\tau > 0$, $G_0 > 0$ and $ga \geq 0$; other parameters require separate analysis. Its delayed state changes phase without asserting the physical identity of x. The measured impedance is $Z = 1/Y$, not Y itself.

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The same current can mix different carriers

Under the independent-carrier approximation, $\sigma = \sum_i n_i |z_i| q \mu_i$ when μ_i is defined as the speed per electric-field magnitude for that carrier. The effective response can include electrons, holes and ions.

For doubly positive oxygen vacancies with concentration c_V and mobility μ_V, $\sigma_V = 2 q c_V \mu_V$. A different mobility convention changes the written charge factor; always state the definition.

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Work a vacancy-conductivity estimate

Assume only vacancy conduction, $c_V = 10^{18}\,\mathrm{cm^{-3}}$, $\mu_V = 8.6 \times 10^{-14}\,\mathrm{cm^2/(V\,s)}$ and $q = 1.602 \times 10^{-19}\,\mathrm{C}$.

Then $\sigma_V = 2 q c_V \mu_V \approx 2.76 \times 10^{-14}\,\mathrm{S/cm}$. A slab with $L = 0.005\,\mathrm{cm}$ and $A = 1\,\mathrm{cm^2}$ has ideal bulk resistance $L/(\sigma_V A) \approx 1.81 \times 10^{11}\,\Omega$. Contact resistance and other carriers are excluded.

For a purely illustrative comparison with $2.2 \times 10^{7}\,\Omega$, the ratio is approximately 8.2×10^{3}; the two rounded values and their conditions must be checked before comparing experimental states.

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What isotope exchange adds

In one reported exchange experiment, ^{18}O was applied at 440 °C, 200 mbar for 1080 min, followed by a depth-sensitive SIMS measurement. A profile is interpreted using a transport and surface-exchange model, including interfaces and the substrate boundary. These numbers belong to this specimen and protocol.

SIMS samples after exposure and depth profiling; it is not automatically an in situ movie of one vacancy. Sputter mixing, depth calibration, surface exchange and non-unique parameter fits contribute uncertainty.

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Relate tracer diffusion and ionic conductivity carefully

In a stated model with tracer diffusion D^* and Haven ratio H, use $D_\sigma = D^*/H$. For dilute independent carriers of charge zq and concentration c, a Nernst–Einstein form is

$$\sigma_{ion} = \frac{D_\sigma c (zq)^2}{k_B T}.$$

The concentration, correlation convention and thermodynamic assumptions must be specified. Isotope tracer motion and electrical transport need not be identical in a correlated or nonideal material.

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A conditional diffusion-to-conductivity example

Suppose $D^* = 6.9 \times 10^{-12}\,\mathrm{cm^2/s}$, $H = 0.69$, $c = 10^{18}\,\mathrm{cm^{-3}}$, $z = 2$, $T = 700\,\mathrm{K}$, and $k_B = 1.381 \times 10^{-23}\,\mathrm{J/K}$. Then $D_\sigma = 1.0 \times 10^{-11}\,\mathrm{cm^2/s}$.

Substitution gives $\sigma_{ion} \approx 1.06 \times 10^{-10}\,\mathrm{S/cm}$. This teaching calculation assumes the stated Nernst–Einstein model and does not infer a concentration from the tracer measurement alone.

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Temperature can reveal a change of regime

If one activated process dominates over a stated temperature interval, $\sigma T = C \exp[-E_a/(k_B T)]$ is one possible analysis form; $\ln(\sigma T)$ versus $1/T$ then has slope $-E_a/k_B$.

The reference reports apparent activation energies of about 1.6 eV above and 0.8 eV below a crossover near 390 °C for a particular film. Its proposed electronic and ionic interpretation is sample- and model-dependent. A slope change alone does not prove which carrier dominates or establish universal values.

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Sweep rate changes the experiment

Let τ_{int} be an internal relaxation time and t_{sweep} the time over which bias changes appreciably. If $t_{sweep} \ll \tau_{int}$, a chosen state variable may be approximately frozen during the sweep. If t_{sweep} is comparable or longer, it may evolve.

This is a comparison of time scales, not a promise that a fast sweep is non-disturbing. Voltage amplitude, heating, read history and other faster processes still matter.

In the reported film study, a state was first established by a fixed bias and then read with a fast sweep; it did not persist beyond one cycle. A negative sweep at 1 V/s still showed hysteresis. These observations belong to the reported setup and do not define a universal rate threshold.

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Separate observation from a film interpretation

For one room-temperature Pt/Fe:STO/Nb:STO film device, reported values under a ± 3 V sweep at 100 mV/s were $R_{ON} = 28\,\mathrm{k\Omega}$ and $R_{OFF} = 150\,\mathrm{k\Omega}$, giving $R_{OFF}/R_{ON} = 150/28 \approx 5.36$. Read conditions and history must also be specified when comparing another experiment.

Another experimental series varied top-electrode material and size and found incomplete repeatability. These sample-specific results motivate reporting the population and protocol, not a guaranteed yield or switching ratio for STO generally.

For instance, 16 repeatable devices out of 40 tested correspond to $16/40 = 40\%$ in that sample. The uncertainty and selection protocol matter; this sample fraction is not a population guarantee.

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Single-crystal defect chemistry needs a model

Acceptor and donor dopants affect charge balance, but an acceptor need not always be compensated only by oxygen vacancies; electrons, holes, other defects and oxygen activity can matter.

Quenching can freeze some defect populations while carrier and interface distributions still evolve. To infer a vacancy concentration from conductivity, state the assumed charged species, mobility and negligible alternatives.

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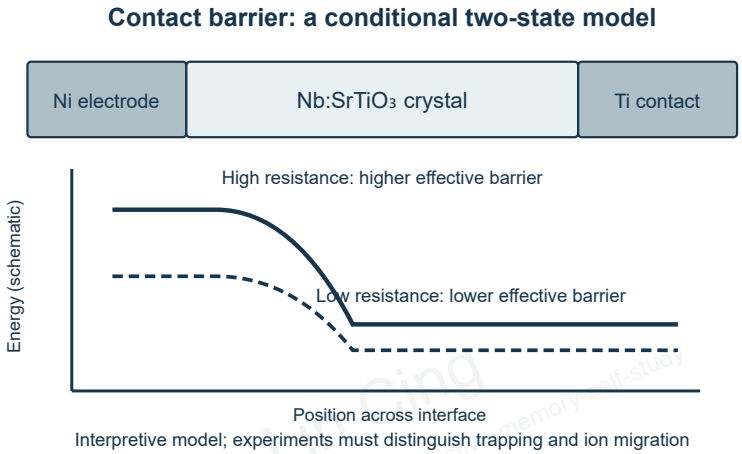
Forming and state redistribution

In one acceptor-doped single-crystal interpretation, formation under a specified field and temperature produces a highly conductive region and a gap near an electrode. Switching is modeled through changes in that local region.

The proposed internal geometry is a model supported by the experiment, not a universal direction or a direct proof from one impedance arc. A single terminal resistance cannot locate all current paths.

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A Schottky contact is a distinct hypothesis



The ideal Schottky–Mott estimate for an n-type contact is $\Phi_B \approx \Phi_M - \chi$ in electron-volts when both terms are given in eV. Interface states, reactions, image-force effects and nonuniform barriers can invalidate the ideal estimate.

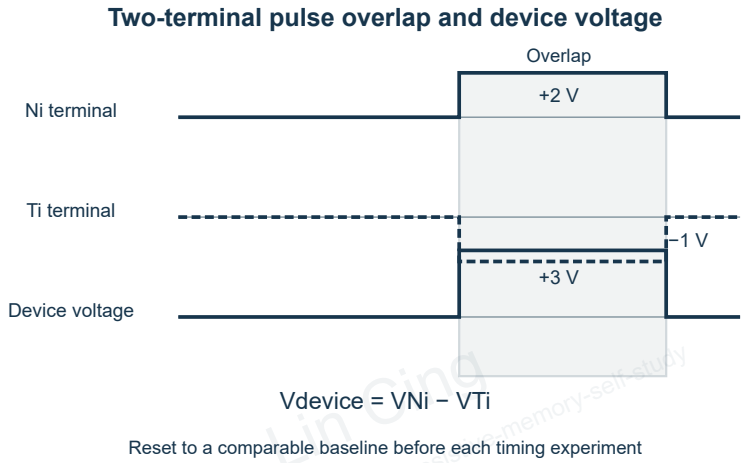
In a reported Ni/Nb:STO/Ti sample, barrier fits differed between resistance states; trapping and oxygen redistribution were both discussed. A fitted barrier is not a direct atom-by-atom image.

Work a restricted barrier comparison

Assume thermionic emission with the same effective area, temperature, ideality factor and reduced bias for two states. In the model, $I \propto \exp[-\Phi_B/(k_B T)]$.

At $T = 300\,\mathrm{K}$, $k_B T \approx 0.02585\,\mathrm{eV}$; lowering Φ_B from 0.72 to $0.50\,\mathrm{eV}$ gives $I_L/I_H = \exp(0.22/0.02585) \approx 4.97 \times 10^3$. This is a hypothetical equal-parameter comparison: the reference's fitted ideality factors differ, so it is not its measured switching ratio.

Pulses are applied between two terminals



Define $V_{device}(t) = V_{Ni}(t) - V_{Ti}(t)$ for the chosen terminal labels. Two waveforms may overlap in time but the device sees their signed difference. Set pulse width, spacing, baseline, read condition and initial state before comparing updates.

Trace two simultaneous pulses

If $V_{Ni} = +2\,\mathrm{V}$ and $V_{Ti} = -1\,\mathrm{V}$, the instantaneous device voltage is $+3\,\mathrm{V}$. If both terminals are at $+2\,\mathrm{V}$, the difference is $0\,\mathrm{V}$.

The update cannot be inferred solely from peak voltage if pulse duration, heating or threshold variability matter. The example does not imply a universal programming threshold.

Pair timing with a controlled initial state

Spike-timing-dependent plasticity (STDP) describes a timing-dependent conductance update. Define $\Delta t = t_{post} - t_{pre}$, both terminal waveforms and the sign convention. Compute $V_{device}(t)$ over the full overlap before associating an observed change with Δt.

Reset to a stated baseline before each timing trial, or model history explicitly. A normalized update also needs its denominator and the read protocol; different normalizations are not interchangeable.

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Relearning is an operational comparison

Suppose reaching a fixed conductance criterion first requires 62 identical pulses, and after a stated delay relearning requires 16. The pulse-count reduction is $1 - 16/62 \approx 74.2\%$.

It is also a time reduction only if pulse durations, spacing, stopping criterion and intervening delays are comparable. A device behavior can resemble a biological learning curve without reproducing the underlying biology.

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Experimental controls for a convincing comparison

Record stack, electrode area, temperature, atmosphere, forming and voltage polarity. Repeat across devices and cycles; include a reference region or control contact. Measure both polarities and multiple sweep rates, then test competing interface and bulk models against spatial and chemical observations.

Report fitted parameters and uncertainty with the fitting window. State whether retention was checked at zero bias after a defined wait; a hysteresis loop alone does not establish long-term retention.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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Resistive switching meets photonics

Electrical switching can change the optical response of a waveguide or resonator. This unit separates the measured electrical state, optical transmission, the physical mechanism proposed for their coupling, and retention after the voltage is removed.

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What is a plasmonic mode?

A surface plasmon polariton (SPP) is an electromagnetic mode coupled to collective charge motion at a conductor–dielectric interface. Its field can be confined near the interface, allowing a compact active region to interact with light.

Confinement is not free: the field penetrating a lossy conductor dissipates power. The best device balances overlap, attenuation, mode conversion, bandwidth and fabrication constraints for a specified geometry and wavelength.

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A planar interface as a model

For two homogeneous, isotropic nonmagnetic half-spaces with complex relative permittivities ϵ_m and ϵ_d, the ideal planar-interface transverse-magnetic (TM) surface mode has

$$\beta = k_0 \sqrt{\frac{\epsilon_m \epsilon_d}{\epsilon_m + \epsilon_d}}, \qquad k_0 = \frac{2\pi}{\lambda_0}.$$

Choose a decaying field branch on each side. A practical metal has complex ϵ_m, so β is generally complex: tighter confinement and higher loss often compete. This formula is not the dispersion relation of every finite, multilayer or patterned structure.

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Deriving the TM boundary condition

Put the interface at $z = 0$, dielectric in $z > 0$ and metal in $z < 0$. For propagation along x with the $e^{-i\omega t}$ convention, use a TM field with $H_y \propto e^{i\beta x - \kappa_d z}$ above and $H_y \propto e^{i\beta x + \kappa_m z}$ below. The physical surface-mode branches satisfy $\operatorname{Re}\kappa_d > 0$ and $\operatorname{Re}\kappa_m > 0$.

In each homogeneous half-space Maxwell's equations give $\kappa_j^2 = \beta^2 - \epsilon_j k_0^2$. Continuity of H_y and tangential E_x, using $\nabla \times H = -i\omega\epsilon_0\epsilon_j E$, yields

$$\frac{\kappa_d}{\epsilon_d} + \frac{\kappa_m}{\epsilon_m} = 0.$$

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Solve, then check the branch

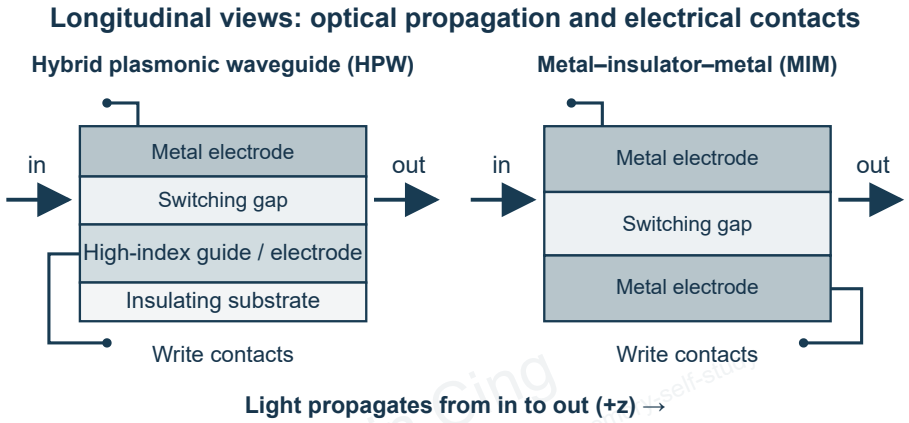
Substituting $\kappa_j^2 = \beta^2 - \epsilon_j k_0^2$ into the squared boundary equation gives

$$\beta^2 = k_0^2 \frac{\epsilon_m \epsilon_d}{\epsilon_m + \epsilon_d}.$$

Squaring can introduce an extraneous solution: substitute the selected square roots back into the **unsquared** boundary condition and verify decay on both sides. For the ideal lossless case with $\epsilon_d > 0$, a bound solution requires $\epsilon_m < -\epsilon_d$. A lossy, finite-thickness or MIM guide needs its own complex dispersion and physical branch selection.

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Two waveguide families



HPW example: conductive high-index guide (e.g., ITO); schematic.

A hybrid plasmonic waveguide (HPW) uses a narrow low-index gap between a high-index guide and metal; a metal–insulator–metal (MIM) structure supports gap-confined modes between conductors. Geometry and surrounding materials determine the mode; the sketches are conceptual longitudinal views, not fabrication recipes.

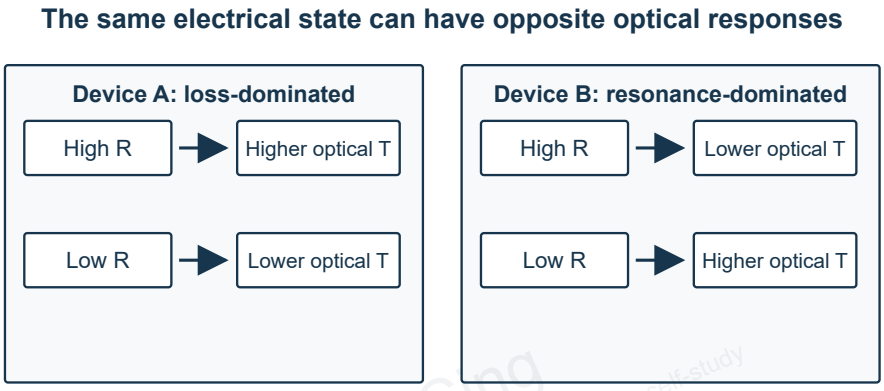
How a resistive state can alter light

The active region may change local complex permittivity, absorption, scattering, carrier density, boundary geometry or a resonance. These effects can coexist. Electrical and optical changes occurring together support a coupling hypothesis but do not alone identify the atoms or defects responsible.

Compare measurements at matched wavelength, input power, polarization and temperature, with a stated reference for normalization. Monitor both transmitted and reflected or scattered light when possible.

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Electrical labels do not determine optical labels



Optical transmission depends on the mode and material, not on resistance alone.

One structure may have lower transmission in its low-resistance state because of increased absorption; another can have higher transmission because a conductive bridge changes its resonance. LRS and HRS are electrical labels. Their optical ordering is a measured property of a particular device, wavelength and geometry.

From a complex effective index to power

Assume a single forward mode, fixed coupling, no reflection or conversion, and the time convention $E \propto e^{i\beta z - i\omega t}$ with $n_{\mathrm{eff}} = n + i\kappa$, $\kappa \geq 0$. Then $\beta = k_0(n + i\kappa)$ and the field magnitude falls as $e^{-k_0 \kappa L}$.

Since power is proportional to squared field magnitude under this normalization,

$$T = \frac{P(L)}{P(0)} = \exp\left(-\frac{4\pi\kappa L}{\lambda_0}\right).$$

Here κ belongs to the **mode**, not automatically to a bulk-material index. Reflections, mode mismatch, scattering and state-dependent coupling require a fuller model.

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A numerical two-state attenuation example

Let $\lambda_0 = 1.55\,\mu\mathrm{m}$ and $L = 2\,\mu\mathrm{m}$. For hypothetical $\kappa_H = 0.01$ and $\kappa_L = 0.08$, the exponents are -0.1621 and -1.2972; therefore $T_H \approx 0.8503$ and $T_L \approx 0.2733$.

These values are conditional on the single-mode assumptions. They do not describe an experimentally measured component or include coupling loss.

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Extinction ratio and insertion loss

With the same input power in both states, define $ER = 10\log_{10}(T_H/T_L)$ and high-transmission-state insertion loss $IL_H = -10\log_{10} T_H$; state the reference plane and any coupling losses.

For the preceding example, $ER \approx 4.929\,\mathrm{dB}$ and $IL_H \approx 0.704\,\mathrm{dB}$. Large extinction does not by itself guarantee low insertion loss or low switching energy.

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Cascading ideal stages

Three identical independent stages with unchanged coupling have $T_{\mathrm{total}} = T_H^3 \approx 0.615$ and $IL_{\mathrm{total}} = 3IL_H \approx 2.112\,\mathrm{dB}$. Losses add in dB because linear transmission factors multiply.

Reflections, coherent interference, amplifier noise and extra couplers invalidate this simple multiplication. Measure system loss at consistent input and output reference planes.

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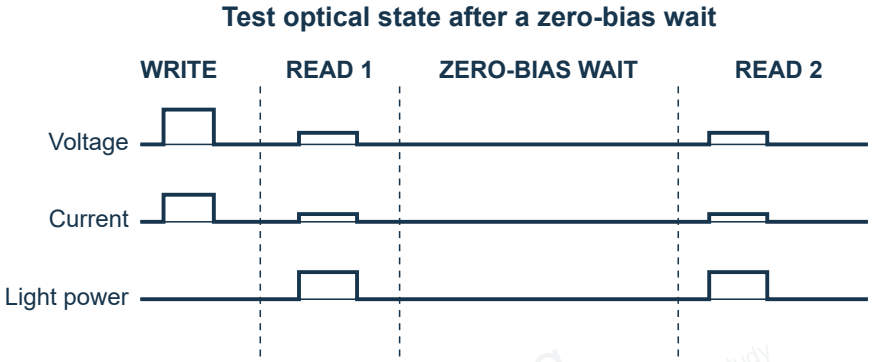
Percentages and decibels are different questions

Suppose a reflection changes from $R_1 = 0.20$ to $R_2 = 0.25$. The absolute change is five percentage points. The ratio is $R_2/R_1 = 1.25$, giving $10\log_{10}(1.25) \approx 0.969\,\mathrm{dB}$.

If only the five-point difference is known, the dB ratio is not determined: a different baseline produces a different ratio. Specify whether reflectance, transmission or a loss is reported.

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Separate writing, reading and waiting



Define the voltage at the device, current, optical probe, timing reference and read disturbance. A practical sequence is write → low-disturbance optical read → remove the bias → wait at controlled temperature → read again.

A hysteresis loop or optical contrast while voltage is applied does not establish zero-bias nonvolatile retention. Repeat after specified wait times and across devices and cycles.

Electrical energy is an integral

Electrical energy delivered to the device over a chosen interval is $E_{\mathrm{dev}} = \int_{t_0}^{t_1} V_{\mathrm{dev}}(t) I_{\mathrm{dev}}(t)\, dt$. Define current direction and measure at the device, since leads and parasitic capacitors can add loss or stored energy.

For an ideal rectangular pulse, $V = 1.2\,\mathrm{V}$, $I = 10\,\mu\mathrm{A}$ and $\Delta t = 20\,\mathrm{ns}$ give $E_{\mathrm{dev}} = 240\,\mathrm{fJ}$. Drivers, reset, sensing, lasers and interconnect can increase full-system energy.

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Power divided by rate needs assumptions

If the average power over a complete observation period, including active and idle intervals, is $P_{\mathrm{avg}} = 12.5\,\mathrm{nW}$ and the average rate is one operation per period at $f = 1\,\mathrm{MHz}$, then $P_{\mathrm{avg}}/f = 12.5\,\mathrm{fJ}$ per operation on average. Instantaneous power need not be constant; the duration, power measurement boundary and definition of an operation matter.

Do not substitute peak power, active-only power or an unrelated maximum frequency into P/f and call it a measured write energy. The average includes whatever idle energy lies within the stated boundary and need not equal one individual write pulse's energy. Integrate synchronized waveforms to distinguish them.

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Historic example: hybrid waveguide

A historical ITO/SiO_2/Au hybrid-waveguide example in the reference reports a roughly $10\ \mu\mathrm{m}$ active length, operation around $\pm 2\ \mathrm{V}$, optical contrast near $12\ \mathrm{dB}$ and a $30\ \mathrm{MHz}$ figure. These belong to a particular device and test configuration, not universal specifications.

The reported $200\ \mathrm{nW}$ is not automatically an energy per operation. Filament composition and VCM/ECM contributions require evidence beyond concurrent current and optical changes.

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A thermal device can be volatile

The reference discusses a VO_2 waveguide in which heating changes the optical state. A thermal mechanism can return toward its original state after electrical power is removed; a sustained bias or other energy input may be necessary.

The reference's body and figure description differ in stated device lengths. Its figure description distinguishes about $1\ \mu\mathrm{m}$ with $12\ \mathrm{dB}$ and $0.5\ \mu\mathrm{m}$ with $4\ \mathrm{dB}$; these are separate devices, not a single combined performance specification.

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Optical and chemical evidence have limits

A ZnO-related grating example in the reference reports a change in optical reflection along with microscopy and elemental analysis consistent with redistribution of oxygen. Contrast and composition measurements support an interpretation, but do not directly map every oxygen vacancy or exclude all alternative changes.

The reported optical percentage is not interchangeable with dB unless both compared reference levels and the measurement definition are known. Retain uncertainty and the illumination geometry.

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A historic a-Si waveguide: optical decrease

An Ag/amorphous-Si hybrid-waveguide example reports a low-resistance state associated with greater absorption and lower optical transmission during a slow sweep from -3 to $+9\,\mathrm{V}$ with approximately $1.5\,\mu\mathrm{m}$ light; the reported modulation is approximately 1%. The electrical and optical traces are observations under that sweep.

The observation alone does not show how long the optical state persists after voltage removal. A matched write–read–zero-bias-wait protocol is needed to characterize retention.

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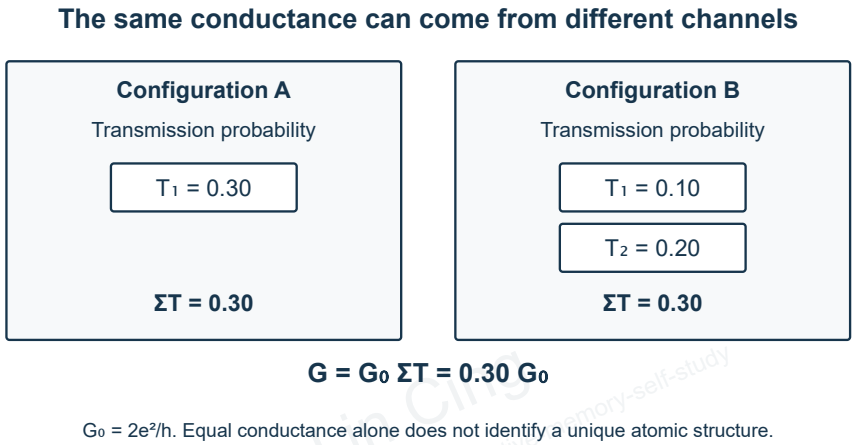
An atomic-scale switch: opposite optical ordering

Another Ag/a-Si MIM configuration reports increased optical transmission in an electrically conducting state. The reference describes an approximately $20\,\mathrm{nm}$ gap and a $1\,\mathrm{kHz}$ sweep with SET near $+1.25\,\mathrm{V}$ and RESET near $+0.25\,\mathrm{V}$ under its bias convention. Its mode and resonant response differ from the previous waveguide, illustrating why an electrical label cannot predict optical sign.

The original study reports $9.2\,\mathrm{dB}$ contrast under its stated conditions (Nano Letters, DOI 10.1021/acs.nanolett.5b04537). The reference gives differing contrast figures in its text and table; use the primary paper's figure with its conditions rather than merging them into an averaged specification. A sweep or biased hold does not by itself establish a particular zero-bias retention time.

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What a conductance step can—and cannot—show



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At low temperature and small bias in an ideal coherent transport model,

$$G = G_0 \sum_n T_n, \qquad G_0 = \frac{2e^2}{h} \approx 77.5\,\mu\mathrm{S},$$

Here $0 \le T_n \le 1$ are channel transmission probabilities.

A conductance calculation and its limits

For one hypothetical channel with $T_1 = 0.3$, $G = 0.3G_0 = 23.25\,\mu\mathrm{S}$ and $R = 1/G \approx 43.0\,\mathrm{k\Omega}$. Two channels with $T_1 = 0.1, T_2 = 0.2$ have the same total in this ideal model. A step can support a small-channel interpretation, but is not a unique atomic photograph.

Finite temperature, contact resistance, inelastic processes, bias dependence and optical field redistribution complicate inference from a measured conductance plateau.

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Quantify a no-failure observation honestly

Suppose 100 independent trials under the **same stated retention time and conditions** show zero failures. For a common failure probability p, $P(0\ \mathrm{failures}) = (1-p)^{100}$.

Setting this probability to 0.05 gives the one-sided 95% upper bound $p_{\mathrm{upper}} = 1 - 0.05^{1/100} \approx 0.0295$, or 2.95%. The bound relies on independent, identically distributed trials: correlated cycles or devices violate the calculation. It says nothing about untested wait times or conditions.

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How to compare reports

For every number, record the structure, state definition, wavelength, polarization, bias, pulse shape, temperature, bandwidth, number of samples and optical reference. Distinguish a stated pulse-integrated energy from a power, and distinguish zero-bias wait from a biased hold.

Optical contrast, switching speed, endurance and retention measured on different devices or in different test runs cannot automatically be combined into one guaranteed operating point.

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Design a discriminating experiment

Use matched electrical and optical time traces with known parasitics. Compare polarization and wavelength dependence, contact materials and controls; probe temperature and zero-bias retention; combine optical data with spatial or chemical measurements when feasible.

If several mechanisms fit the same traces, report the competing explanations and what observation could distinguish them. A correlation or a fitted curve alone rarely determines a unique microscopic mechanism.

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References and attribution

Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

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